CCE107L INTERMEDIATE PROGRAMMING

Wibrary
Mini Library System

An Undergraduate Course Requirement

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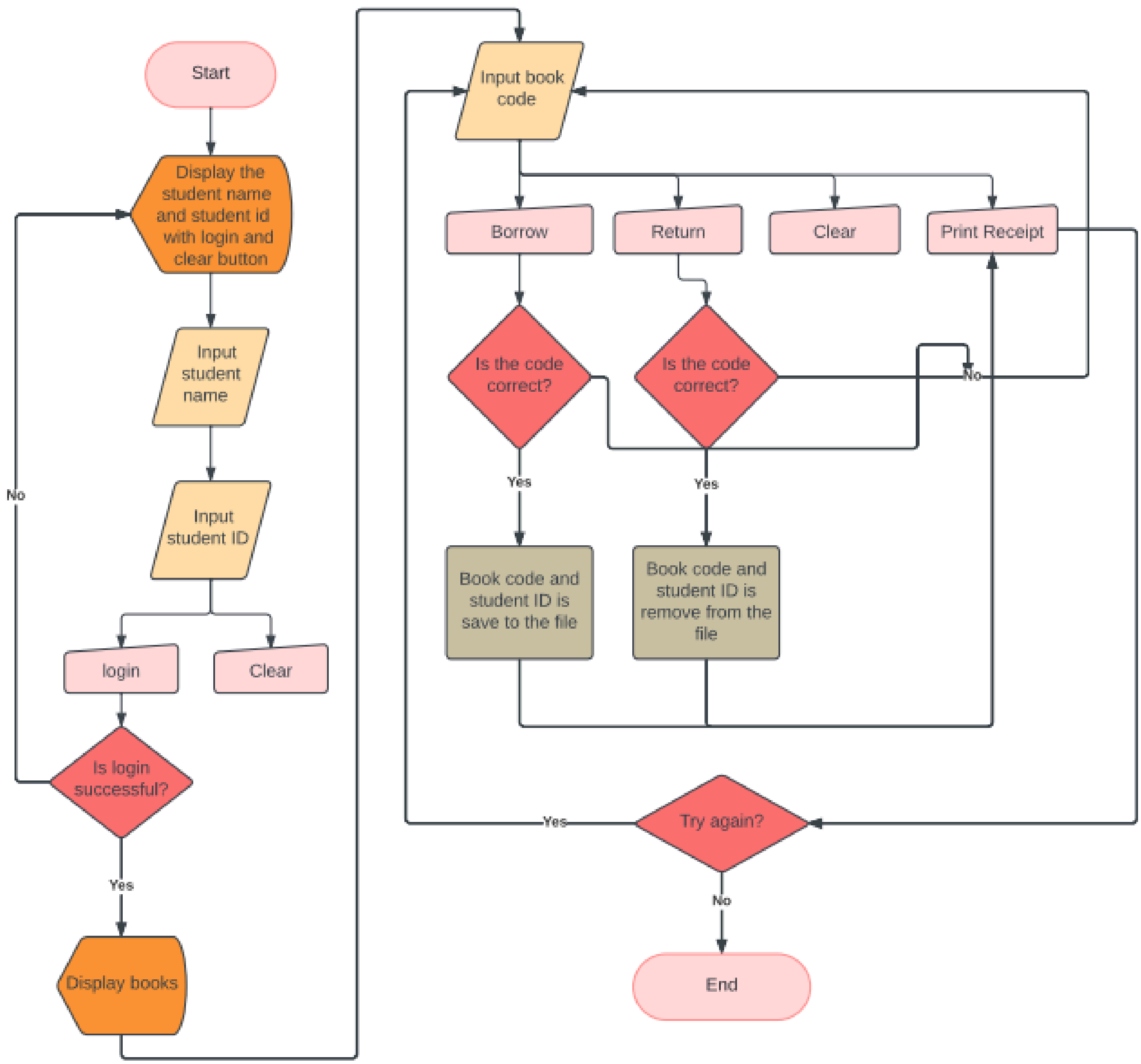
I. Introduction

Wibrary is a mini library management system consisting of programming books. The system allows students to borrow and return books by entering their student name and ID. The programmer visualized it as a kiosk in a library to simplify the process of borrowing and returning books, making it more convenient for students and librarians since it manually tracks the borrowed and returned books.

The first interface, which is for login authentication, prompts students to enter their full name and student ID. If they log in successfully, the second interface will pop up, displaying a list of available programming books. Students can then enter the book code and choose to borrow or return it by clicking the corresponding button. Students can borrow a maximum of three books at a time, with a deadline to return them in seven days and a late fee of 10 pesos will be charged for each day past the deadline. They need to return borrowed books before being able to borrow again. A message will be displayed indicating the success or failure of the operation, and a receipt can be printed by clicking the "Print" button.

II. Flowchart



III. Data Types

A. 1ST CLASS FOR BOOKS

Boolean – The Boolean Data Type is used in a conditional statement in actionPerformed method of some JButtons in order to control the flow and interactivity of JButtons.

Void - Is a Java keyword. Used at method declaration and definition to specify that the method does not return any type, the method returns void.

String - Is an object that represents a number of character values. Each letter in the string is a separate character value that makes up the Java string object.

Protected - Is an access modifier used for attributes, methods and constructors, making them accessible in the same package and subclasses.

This - Is a reference variable that refers to the current object.

Return - Is a reserved keyword i.e., we can't use it as an identifier. It is used to exit from a method, with or without a value.

B. 2ND CLASS FOR STUDENTS

Protected - Is an access modifier used for attributes, methods and constructors, making them accessible in the same package and subclasses.

Int – The int Data type is used in our program since they are used for the measurement and location of the frame which require numerical value. Variables with int Data Types can also be used in initializing the value of JSpinners and JTextFields.

String - Is an object that represents a number of character values. Each letter in the string is a separate character value that makes up the Java string object.

Void - Is a Java keyword. Used at method declaration and definition to specify that the method does not return any type, the method returns void.

Boolean – The Boolean Data Type is used in a conditional statement in actionPerformed method of some JButtons in order to control the flow and interactivity of JButtons.

C. 3RD CLASS RECEIPT

BufferedWriter - is used to provide buffering for Writer instances. It makes the performance fast. It inherits Writer class.

FileWriter - is a specialized OutputStreamWriter for writing character files.

IOException - is a checked exception that must be handled at compilation time. IOException is the base class for such exceptions which are thrown while accessing data from files, directories and streams.

DateFormat - is an abstract class for date/time formatting subclasses which formats and parses dates or time in a language-independent manner.

SimpleDateFormat - is a concrete class for formatting and parsing dates in a locale-sensitive manner.

Date - The class Date represents a specific instant in time, with millisecond precision. The Date class of java.util package implements.

String - Is an object that represents a number of character values. Each letter in the string is a separate character value that makes up the Java string object.

This - Is a reference variable that refers to the current object.

Try Catch - The try statement allows you to define a block of code to be tested for errors while it is being executed.

Writer Class - The write (String) method of Writer Class in Java is used to write the specified String on the stream.

D. 4th CLASS "LIBRARY"

ArrayList - The ArrayList class is a resizable array, which can be found in the java.util package.

HashMap - in Java stores the data in (Key, Value) pairs, and you can access them by an index of another type (e.g. an Integer).

Map - Map Interface is present in java.util package represents a mapping between a key and a value. Java Map interface is not a subtype.

Try Catch - The try statement allows you to define a block of code to be tested for errors while it is being executed.

String - Is an object that represents a number of character values. Each letter in the string is a separate character value that makes up the Java string object.

While loop - While loop is a control flow statement that allows code to be executed repeatedly based on a given Boolean condition.

If-else - statement also tests the condition. It executes the *if block* if the condition is true otherwise the *else block* is executed.

For loop - Iterates a given set of statements multiple times. The Java while loop executes a set of instructions until a boolean condition is met.

Return - the return statement is used for returning a value when the execution of the block is completed.

Break - The break statement in java is used to terminate from the loop immediately.

E. LOGIN INTERFACE

JFrame – A class that represents the main window of a GUI application.

JOption – A utility Class in Java that simplifies the process of creating and displaying dialog boxes in a GUI application.

JPanel – A class that represents a container for other components in a GUI application.

initComponents - is a method that NetBeans swing designer creates to initialize components.

String - Is an object that represents a number of character values. Each letter in the string is a separate character value that makes up the Java string object.

JOptionPane - is a class library that makes it easy to pop up a simple dialog box that either provides an information message or asks for a simple input from the user.

JTextField - class is a text component that allows the editing of a single line text. It inherits JTextComponent class.

JButton - The JButton class is used to create a labeled button that has platform independent implementation.

JLabel - is a built-in Java Swing class that lets you display information on a JFrame.

Try Catch - The try statement allows you to define a block of code to be tested for errors while it is being executed.

JTextArea - is a part of java Swing package. It represents a multi-line area that displays text. It is used to edit the text.

IV. Swing Graphical User Interfaces and File Handling

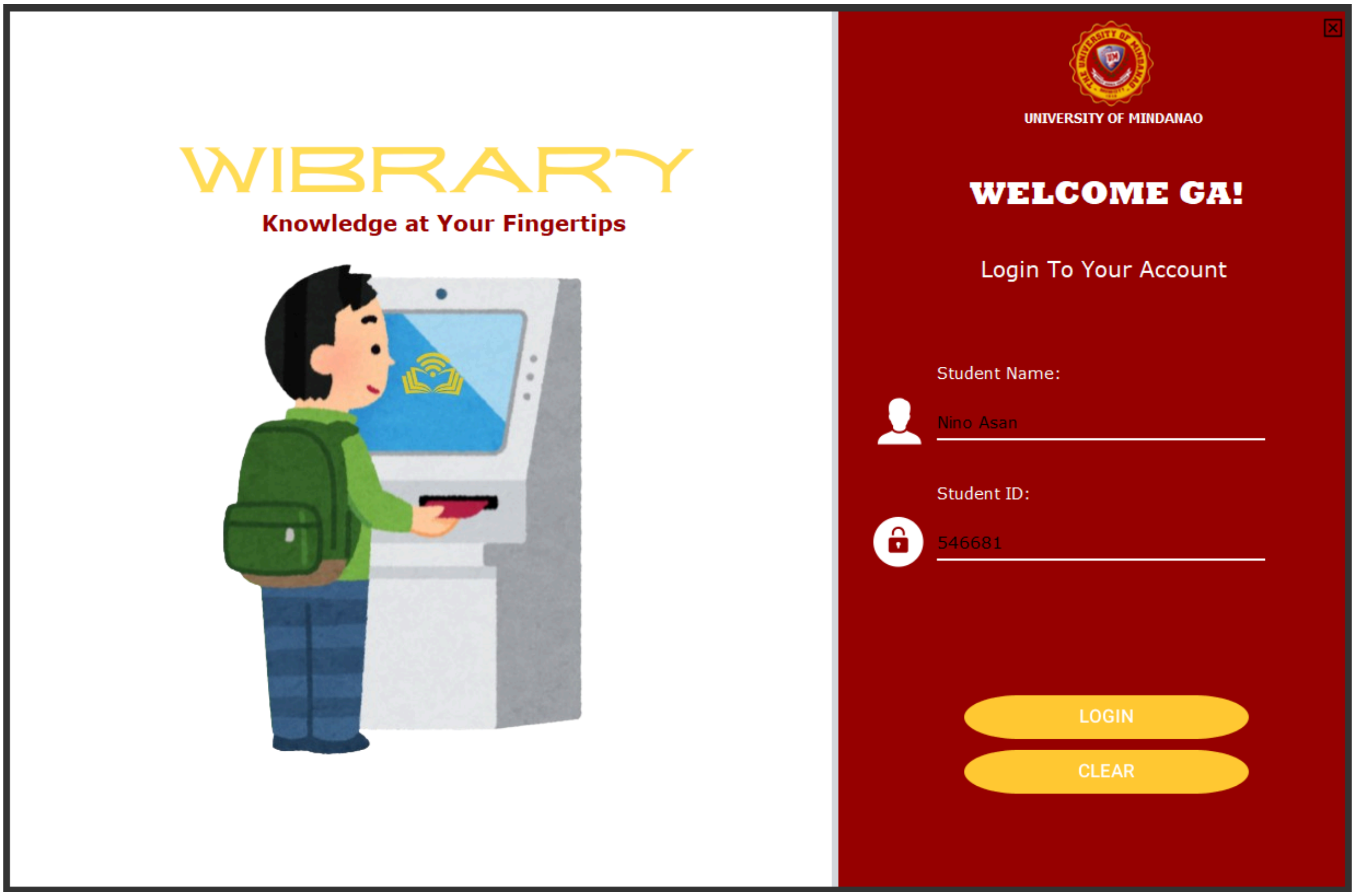


Figure 2. Login Menu
Input name and Student ID given by the school

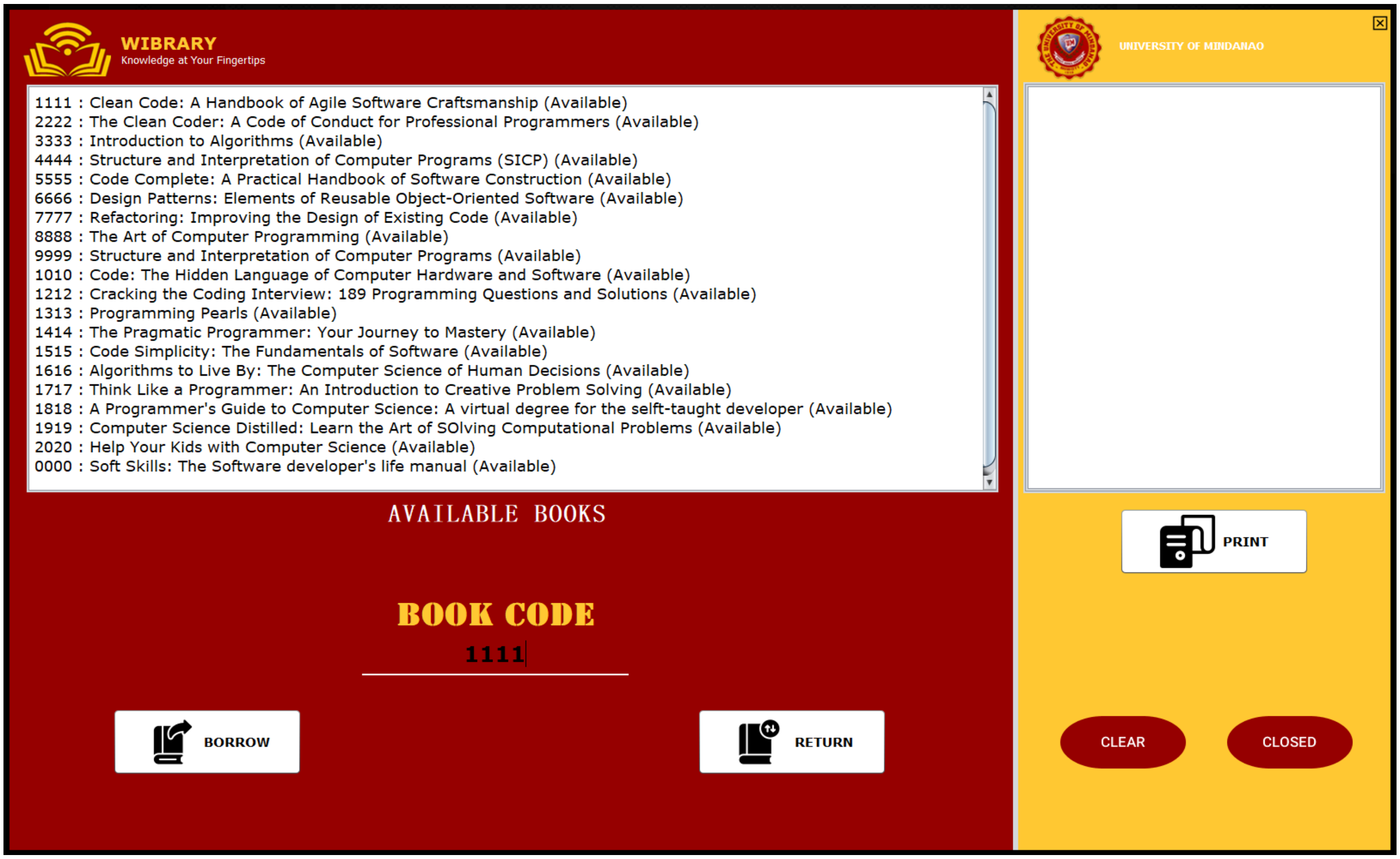


Figure 3. Books and Option Frame

Type the code number of the book

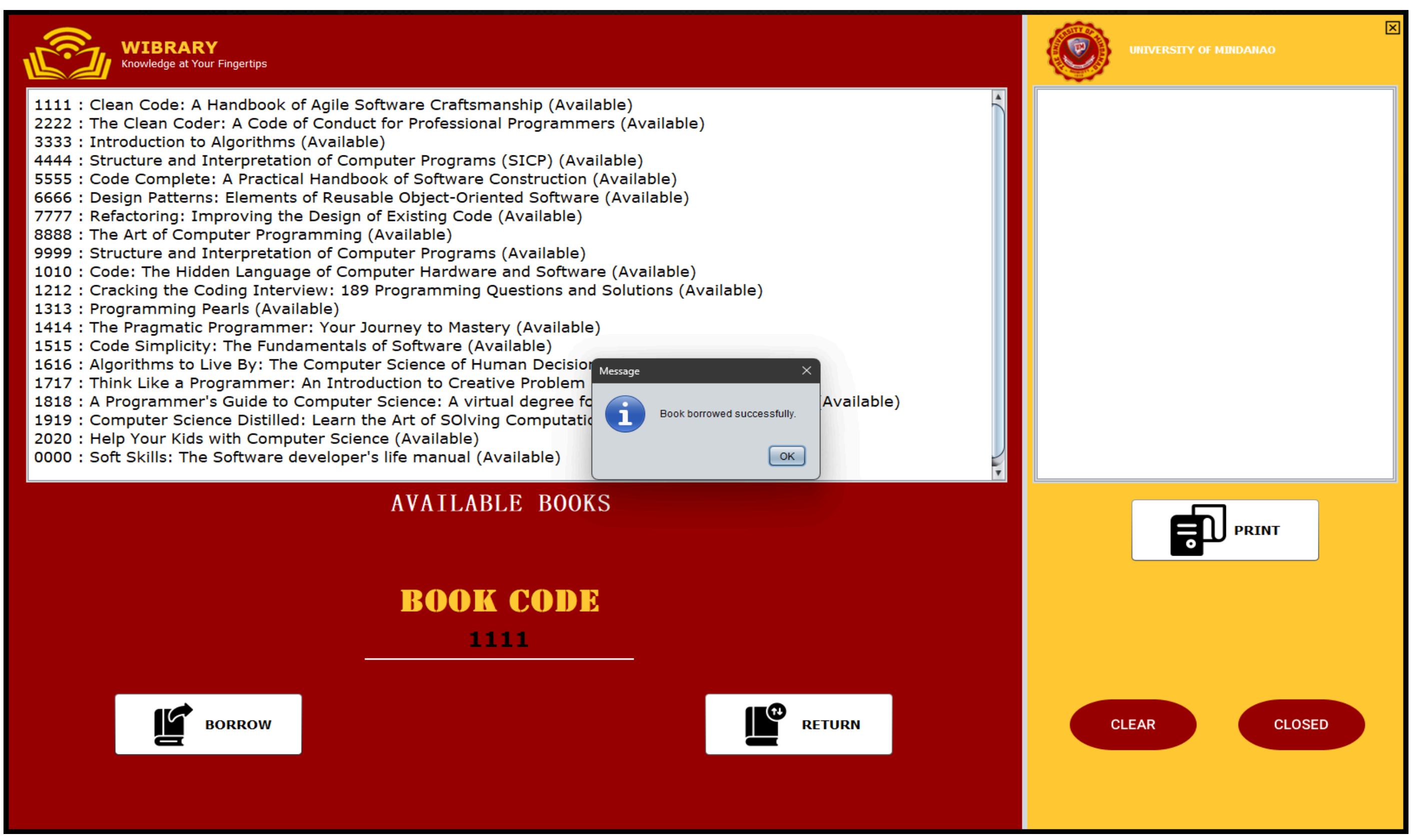


Figure 4. Output Message

Once code is entered you will get a message on whether it’s successful or not.

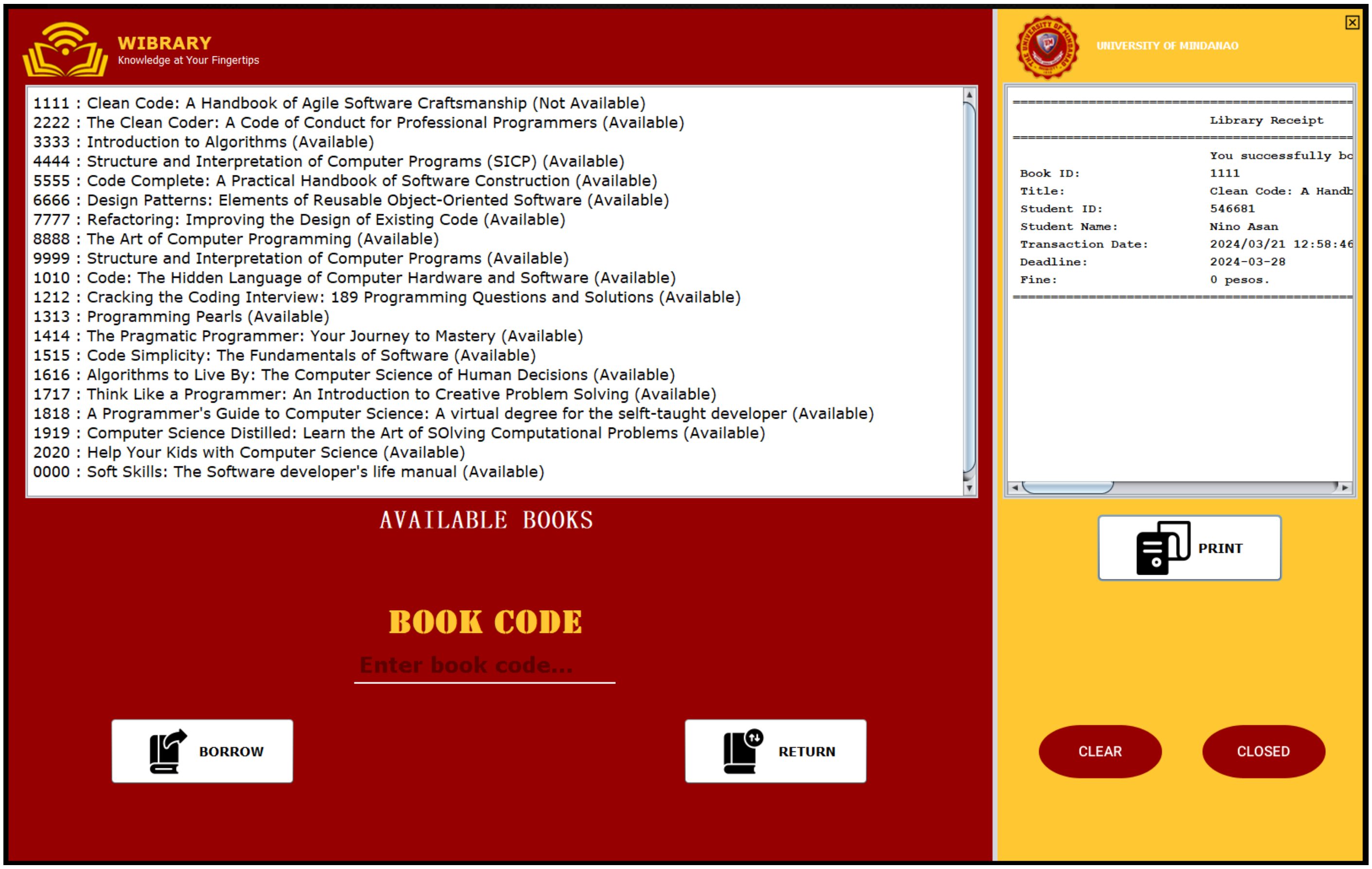


Figure 5. Receipt Frame

After you have successfully completed all requirements you will get a receipt and have the option to print or clear it.

JFrame - is a swing component used to view the incorporated swing GUI components. It is also the component where the JPanel is placed.

JPanel - is a swing component where other swing GUI components except the JFrame are placed. Different panels are being utilized in the program to have different sections which makes it more appealing and organized.

JButton - is a swing component used to create buttons in the program. This component is incorporated with ActionListener to allow certain actions to happen when it is clicked. It is used in the program to save, update, and delete and redirect users to other frames.

JTextField - is a swing component that allows users to input single line texts like sign-up details.

JLabel - is a swing component used in the program to insert texts and images to the frame.

JTable - is a swing component that provides a graphical table component for displaying and editing data in a tabular form.

JTextArea - is a part of java Swing package. It represents a multi-line area that displays text. It is used to edit the text.

JscrollPane - is a combination of viewports and scrollbars. It will connect our viewport with the scrollbar.

V. Learnings

In taking this course, we learned and gained a lot of knowledge from this course Intermediate Programming such as knowing more about the different concepts or methods in programming. We also learned techniques on how to solve our problems in problem solving which helps us make solving easier in the future. We also gained deeper understanding about file handling, Object-Oriented Programming (OOP), GUI, array and more.

While we were developing this Wibrary for our project, we learned and gained more knowledge especially in creating the login form and how to make the borrowing and returning of books function. Also, during developing this project we realized how important planning, organizing, testing and debugging is to make sure the program works. We also realized how important communication and teamwork is. Because we had good communication and worked as a team, we successfully made this project with no problems

Overall, Intermediate Programming can provide students like us improve and prepare us for the future studies or career. Studying this course was a great experience and helped us become successful and better future programmers, because of this successful project we are excited to learn more and gain deeper knowledge about programming that could be really useful and helpful for our future.

VI. Java Program Codes

Class for Book2

```
/*
 * To change this license header, choose License Headers in Project Properties.
 * To change this template file, choose Tools | Templates
 * and open the template in the editor.
 */
package library.kiosk;

import java.io.Serializable;

/**
 *
 * @author Ninz
 */
public class Book2 implements Serializable {
    private static final long serialVersionUID = 1L;
     protected String id;
     protected String title;
     protected boolean available;
     protected String borrowerId;
     public boolean borrowed;

    public Book2(String id, String title) {
     this.id = id;
        this.title = title;
        this.available = true;
        this.borrowerId = null;
    }

      public String getBorrowerId(){
    return borrowerId;
    }

    public void setBorrowerId(String borrowerId){
        this.borrowerId = borrowerId;
    }

    public String getId() {
        return id;
    }

    public String getTitle() {
        return title;
    }
    public String getBook(){
        return getId()+" "+getTitle();
    }

    public boolean isAvailable() {
        return available;
```

```
    }

    public void setAvailable(boolean available) {
        this.available = available;
    }

    public boolean isBorrowed() {
        return borrowed;
    }

    public void setBorrowed(boolean borrowed) {
        this.borrowed = borrowed;
    }
}
```

Class for Library

```
/*
 * To change this license header, choose License Headers in Project Properties.
 * To change this template file, choose Tools | Templates
 * and open the template in the editor.
 */
package library.kiosk;

import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.File;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.HashSet;
import java.util.Map;
import java.util.Set;
import javax.swing.JOptionPane;
import javax.swing.JTextArea;

/**
 *
 * @author Ninz
 */
public class Library {
      public ArrayList<Book2> books = new ArrayList<>(); // BOOKS
    public Map<String, Student2> students = new HashMap<>(); //STUDENTS

     private static final String BORROWER_FILE = "C:\\Program Files\\NetBeansProjects\\file.txt"; //new

     public void loadBorrowersFromFile() {
        try (BufferedReader reader = new BufferedReader(new FileReader(BORROWER_FILE))) {
            String line;
            while ((line = reader.readLine()) != null) {
```

```java
            String[] parts = line.split(",");
            String bookId = parts[0];
            String borrowerId = parts[1];
            // Set borrower ID for the book
            setBorrowerIdForBook(bookId, borrowerId);
            System.out.println("loadBorrowersFromFile successfully.");
        }
    } catch (IOException e) {
        System.err.println("Error loading borrowers from file: " + e.getMessage());
    }
}


// Save borrower ID to file : NEW
private void saveBorrowerToFile(String bookId, String studentId) {
    try (BufferedWriter writer = new BufferedWriter(new FileWriter(BORROWER_FILE, true))) {
        writer.write(bookId + "," + studentId);
        writer.newLine();
    } catch (IOException e) {
        System.err.println("Error saving borrower to file: " + e.getMessage());
    }
}

// Remove borrower ID from file when book is returned : NEW
private void removeBorrowerFromFile(String bookId) {


try {
    File borrowersFile = new File(BORROWER_FILE);
    File tempFile = new File(BORROWER_FILE + ".tmp");
    try (BufferedReader reader = new BufferedReader(new FileReader(borrowersFile));
         PrintWriter writer = new PrintWriter(new FileWriter(tempFile))) {
        String line;
        while ((line = reader.readLine()) != null) {
            if (!line.startsWith(bookId + ",")) {
                writer.println(line);
            }
        }
    }
    // Delete the original file
    if (!borrowersFile.delete()) {
        throw new IOException("Could not delete original file");
    }
    // Rename the temporary file to the original file
    if (!tempFile.renameTo(borrowersFile)) {
        throw new IOException("Could not rename temp file to original file");
    }
} catch (IOException e) {
    System.err.println("Error removing borrower from file: " + e.getMessage());
}

}

// Helper method to set borrower ID for a book :
private void setBorrowerIdForBook(String bookId, String borrowerId) {
    for (Book2 book : books) {
```

```
            if (book.getId().equals(bookId)) {
                book.setBorrowerId(borrowerId);
                break;
            }
        }
    }



    public ArrayList<Book2> getBooks() {
    return books;
}

    public void loadBooksFromFile(String filename) {
        try (BufferedReader br = new BufferedReader(new FileReader("C:\\Program
Files\\NetBeansProjects\\books.txt"))) {
            String line;
            while ((line = br.readLine()) != null) {
                String[] parts = line.split(",");
                String id = parts[0];
                String title = parts[1];
                books.add(new Book2(id, title));
            }
            System.out.println("Books loaded successfully.");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    public void initializeStudents(String filename) { //LIST OF STUDENTS IN THE SCHOOL
        try (BufferedReader br = new BufferedReader(new FileReader("C:\\Program
Files\\NetBeansProjects\\students.txt"))) {
            String line;
            while ((line = br.readLine()) != null) {
                String[] parts = line.split(",");
                if (parts.length == 2) {
                    String studentId = parts[0].trim();
                    String studentName = parts[1].trim();
                    students.put(studentId, new Student2(studentId, studentName));
                } else {
                    System.err.println("Invalid format in file for line: " + line);
                }
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    public void displayBooks2(JTextArea bookTextArea) { //NEW DISPLAY METHOD newwwwww

    // Clear the content of the JTextArea before displaying the book list
    bookTextArea.setText("");

    // Create a Set to keep track of displayed book IDs
    Set<String> displayedBookIds = new HashSet<>();
```

```
    for (Book2 book : books) {
    if (!book.isBorrowed() && book.isAvailable() && !displayedBookIds.contains(book.getId())) {
        bookTextArea.append(book.getId() + " : " + book.getTitle() +" (Available)"+"\n");
    } else if (!book.isAvailable() && !displayedBookIds.contains(book.getId())) {
        bookTextArea.append(book.getId() + " : " + book.getTitle() + " (Not Available)\n");
    }
    displayedBookIds.add(book.getId());
    }
}
    public void displayBooks(JTextArea bookTextArea) {
        for (Book2 book : books) {
            if (book.isAvailable()) {
                bookTextArea.append(book.getId() + " : " + book.getTitle() + "\n");
            }
        }
    }
    public void displayReceipt(JTextArea receipt) {
        try (BufferedReader reader = new BufferedReader(new FileReader("C:\\Program
Files\\NetBeansProjects\\receipt.txt"))) {

            StringBuilder receiptText = new StringBuilder();
            String line;
            while ((line = reader.readLine()) != null) {
                receiptText.append(line).append("\n");
            }
            receipt.setText(receiptText.toString());
        } catch (IOException e) {
            e.printStackTrace();

        }
    }


    public void borrowBook(String bookId, String studentId, String studentName, JTextArea bookTextArea) {


        boolean bookFound = false;
        boolean studentFound = false;

        String message1 = "You successfully borrowed a book.";
        String message2 = "You have reached the maximum limit of borrowed books. Return them before
borrowing again.";




        // Check if the student exists
        if (students.containsKey(studentId)) {
            Student2 student = students.get(studentId);

            // Check if the student has already borrowed 3 books
            if (student.getNumberOfBooksBorrowed() >= 3) {
                JOptionPane.showMessageDialog(null, message2);
                return;
            }
            studentFound = true;
```

```
        } else {
            JOptionPane.showMessageDialog(null, "Invalid student ID.");
            return;
        }


            for (Book2 book : books) {
                if (book.getId().equals(bookId)) {
                    bookFound = true;
                    if (book.isAvailable()) {
                        book.setAvailable(false);
                        book.setBorrowerId(studentId);
                        book.setBorrowed(true);

                        Resibo receipt = new Resibo(message1,book.getId(), book.getTitle(), studentId,
studentName);
                        receipt.saveToFile("C:\\Program Files\\NetBeansProjects\\receipt.txt");

                        // Increment the number of books borrowed by the student
                         Student2 student = students.get(studentId);
                        student.incrementNumberOfBooksBorrowed();

                        //Set the borrower ID for the borrowed book
                        setBorrowerIdForBook(bookId,studentId);
                        // Save borrower ID to file
                        saveBorrowerToFile(bookId, studentId);

                        JOptionPane.showMessageDialog(null, "Book borrowed successfully.");
                        displayBooks2(bookTextArea);
                    } else {
                        JOptionPane.showMessageDialog(null, "Book not available for borrowing.");
                    }
                    break;
                }

            }if(!bookFound){
                    JOptionPane.showMessageDialog(null,"Invalid book ID.");
            }if(!studentFound){
                   JOptionPane.showMessageDialog(null,"Invalid student ID.");

            }
    }




    public void returnBook(String bookId,String studentId,String studentName, JTextArea bookTextArea) {
        boolean bookFound = false;

        String message1 = "You successfully returned a book.";
        String message2 = "You have not borrowed this book.";
        String message3 = "Invalid student ID.";
        String message4 = "Invalid book ID.";
```

```
      // Check if the student exists
   if (!students.containsKey(studentId)) {
      JOptionPane.showMessageDialog(null, message3);
      return;
   }

         // Check if the book exists
   Book2 bookToReturn = null;
   for (Book2 book : books) {
      if (book.getId().equals(bookId)) {
         bookToReturn = book;
         bookFound = true;
         break;
      }
   }
   if (!bookFound) {
      JOptionPane.showMessageDialog(null, message4);
      return;
   }

   // Check if the book is borrowed by the student
   if (!bookToReturn.isAvailable() && bookToReturn.getBorrowerId().equals(studentId)) {
      bookToReturn.setAvailable(true);
      bookToReturn.setBorrowed(false);

               Resibo receipt = new Resibo(message1, bookToReturn.getId(), bookToReturn.getTitle(),
studentId, studentName);
               receipt.saveToFile("C:\\Program Files\\NetBeansProjects\\receipt.txt");
               JOptionPane.showMessageDialog(null, "Book returned successfully.");

               // Decrement the number of books borrowed by the student
               Student2 student = students.get(studentId);
               student.decrementNumberOfBooksBorrowed();

               // Remove borrower ID from file
               removeBorrowerFromFile(bookId);

               displayBooks2(bookTextArea);

             } else {
      JOptionPane.showMessageDialog(null, message2);
   }

   }
}
```

Class for Resibo

```
/*
 * To change this license header, choose License Headers in Project Properties.
 * To change this template file, choose Tools | Templates
 * and open the template in the editor.
 */
package library.kiosk;

import java.io.BufferedWriter;
import java.io.FileWriter;
import java.io.IOException;
import java.text.DateFormat;
import java.text.SimpleDateFormat;
import java.util.Date;

/**
 *
 * @author Ninz
 */
public class Resibo {
    String message;
     String bookId;
     String bookTitle;
     String studentId;
     String studentName;
     Date transactionDate;

    public Resibo(String message,String bookId, String bookTitle, String studentId, String studentName) {
        this.message = message;
        this.bookId = bookId;
        this.bookTitle = bookTitle;
        this.studentId = studentId;
        this.studentName = studentName;
        this.transactionDate = new Date();
    }

    public void saveToFile(String filePath) {
        try (BufferedWriter writer = new BufferedWriter(new FileWriter(filePath, false))) {

            // Clear the old receipt by truncating the file
            writer.write(""); // This effectively clears the file


            DateFormat dateFormat = new SimpleDateFormat("yyyy/MM/dd HH:mm:ss");
            String dateString = dateFormat.format(transactionDate);


writer.write("==================================================================
==========================================\n");
            writer.write("                    Library Receipt                                        \n");

writer.write("==================================================================
==========================================\n");
            writer.write("                    " + message + "                                        \n");
```

```
            writer.write(" Book ID:                  " + bookId + "                                                                   \n");
            writer.write(" Title:                    " + bookTitle + "                                                              \n");
            writer.write(" Student ID:               " + studentId + "                                                                   \n");
            writer.write(" Student Name:             " + studentName + "
\n");
            writer.write(" Transaction Date:         " + dateString + "
\n");
            writer.write(" Deadline:                 " + setDeadline() + "
\n");
            writer.write(" Fine:                     " + calculateFine() + " pesos."+"
\n");

writer.write("==========================================================================
==========================================\n");

            writer.newLine();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

  private int calculateFine() {

        DateFormat dateFormat = new SimpleDateFormat("yyyy-MM-dd");
        Date deadlineDate;
        try {
            deadlineDate = dateFormat.parse(setDeadline());
        } catch (Exception e) {
            e.printStackTrace();
            return 0;
        }


        if (transactionDate.after(deadlineDate)) {

            long difference = transactionDate.getTime() - deadlineDate.getTime();
            int daysLate = (int) (difference / (1000 * 60 * 60 * 24));

            // Apply fine of 10 pesos per day late
            return daysLate * 10;
        } else {
            return 0; // No fine if returned on time
        }
    }

  //METHOD 3.
    private String setDeadline() {
        // Set deadline to 7 days from the borrowing date
        DateFormat dateFormat = new SimpleDateFormat("yyyy-MM-dd");
        Date currentDate = new Date();
        Date deadlineDate = new Date(currentDate.getTime() + (7 * 24 * 60 * 60 * 1000)); // 7 days

        return dateFormat.format(deadlineDate);
    }
}
```

Class for Student2

```
/*
 * To change this license header, choose License Headers in Project Properties.
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 * and open the template in the editor.
 */
package library.kiosk;

/**
 *
 * @author Ninz
 */
public class Student2 {
     protected String id;
    protected String name;
    protected int numberOfBooksBorrowed;

    public Student2(String id, String name) {
        this.id = id;
        this.name = name;
        this.numberOfBooksBorrowed = 0;
    }

    public String getId() {
        return id;
    }

    public String getName() {
        return name;
    }
    public int getNumberOfBooksBorrowed() {
        return numberOfBooksBorrowed;
    }

    public void incrementNumberOfBooksBorrowed() {
        numberOfBooksBorrowed++;
    }

    public void decrementNumberOfBooksBorrowed() {
        if (numberOfBooksBorrowed > 0) {
            numberOfBooksBorrowed--;
        }
    }
    public boolean hasBorrowedBook() {
        return numberOfBooksBorrowed > 0;
    }
}
```

Class for Login Authentication

```
/*
 * To change this license header, choose License Headers in Project Properties.
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 * and open the template in the editor.
 */
package library.kiosk;

import javax.swing.JOptionPane;

/**
 *
 * @author Ninz
 */
public class LoginAuthentication extends javax.swing.JFrame {

    /**
     * Creates new form LoginAuthenticaion
     */
    public LoginAuthentication() {
        initComponents();
    }

    /**
     * This method is called from within the constructor to initialize the form.
     * WARNING: Do NOT modify this code. The content of this method is always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
        jLabel4 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
        jLabel12 = new javax.swing.JLabel();
        jPanel2 = new javax.swing.JPanel();
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jLabel5 = new javax.swing.JLabel();
        jLabel6 = new javax.swing.JLabel();
        jLabel7 = new javax.swing.JLabel();
        jLabel8 = new javax.swing.JLabel();
        jTextField1 = new app.bolivia.swing.JCTextField();
        btnLogin1 = new rojerusan.RSMaterialButtonCircle();
        jLabel9 = new javax.swing.JLabel();
        jTextField2 = new app.bolivia.swing.JCTextField();
        jLabel10 = new javax.swing.JLabel();
        jLabel11 = new javax.swing.JLabel();
        btnLogin2 = new rojerusan.RSMaterialButtonCircle();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setBackground(new java.awt.Color(255, 255, 255));
        setUndecorated(true);
```

```
        jPanel1.setBackground(new java.awt.Color(255, 255, 255));
        jPanel1.setLayout(new org.netbeans.lib.awtextra.AbsoluteLayout());

        jLabel4.setBackground(new java.awt.Color(255, 255, 255));
        jLabel4.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\Untitled design (6).png")); //
NOI18N
        jPanel1.add(jLabel4, new org.netbeans.lib.awtextra.AbsoluteConstraints(80, 210, 470, 490));

        jLabel3.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\WIBRARY.png")); // NOI18N
        jPanel1.add(jLabel3, new org.netbeans.lib.awtextra.AbsoluteConstraints(140, 120, -1, 60));

        jLabel12.setFont(new java.awt.Font("Verdana", 1, 20)); // NOI18N
        jLabel12.setForeground(new java.awt.Color(153, 0, 0));
        jLabel12.setText("Knowledge at Your Fingertips");
        jPanel1.add(jLabel12, new org.netbeans.lib.awtextra.AbsoluteConstraints(230, 180, 340, -1));

        jPanel2.setBackground(new java.awt.Color(153, 0, 0));
        jPanel2.setLayout(new org.netbeans.lib.awtextra.AbsoluteLayout());

        jLabel1.setFont(new java.awt.Font("Tahoma", 1, 12)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(255, 255, 255));
        jLabel1.setText("UNIVERSITY OF MINDANAO");
        jPanel2.add(jLabel1, new org.netbeans.lib.awtextra.AbsoluteConstraints(170, 90, -1, -1));

        jLabel2.setIcon(new javax.swing.ImageIcon("C:\\Program Files\\NetBeansProjects\\Mini Library
Kiosk\\src\\Icon\\imresizer-1709050662631.png")); // NOI18N
        jPanel2.add(jLabel2, new org.netbeans.lib.awtextra.AbsoluteConstraints(190, 0, 100, 90));

        jLabel5.setFont(new java.awt.Font("Tw Cen MT Condensed Extra Bold", 1, 15)); // NOI18N
        jLabel5.setForeground(new java.awt.Color(255, 255, 255));
        jLabel5.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\close-window.png")); //
NOI18N
        jLabel5.addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseClicked(java.awt.event.MouseEvent evt) {
                jLabel5MouseClicked(evt);
            }
        });
        jPanel2.add(jLabel5, new org.netbeans.lib.awtextra.AbsoluteConstraints(440, 0, -1, 30));

        jLabel6.setBackground(new java.awt.Color(255, 255, 255));
        jLabel6.setFont(new java.awt.Font("Rockwell Extra Bold", 0, 30)); // NOI18N
        jLabel6.setForeground(new java.awt.Color(255, 255, 255));
        jLabel6.setText("WELCOME GA!");
        jPanel2.add(jLabel6, new org.netbeans.lib.awtextra.AbsoluteConstraints(120, 140, 270, 50));

        jLabel7.setFont(new java.awt.Font("Verdana", 0, 20)); // NOI18N
        jLabel7.setForeground(new java.awt.Color(255, 255, 255));
        jLabel7.setText("Login To Your Account");
        jPanel2.add(jLabel7, new org.netbeans.lib.awtextra.AbsoluteConstraints(130, 220, -1, 30));

        jLabel8.setFont(new java.awt.Font("Verdana", 0, 15)); // NOI18N
        jLabel8.setForeground(new java.awt.Color(255, 255, 255));
        jLabel8.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\icons8_Secure_50px.png"));
// NOI18N
        jPanel2.add(jLabel8, new org.netbeans.lib.awtextra.AbsoluteConstraints(30, 460, 50, 50));
```

```
        jTextField1.setBackground(new java.awt.Color(153, 0, 0));
        jTextField1.setBorder(javax.swing.BorderFactory.createMatteBorder(0, 0, 2, 0, new java.awt.Color(255,
255, 255)));
        jTextField1.setFont(new java.awt.Font("Verdana", 0, 15)); // NOI18N
        jTextField1.setPlaceholder("Enter Student Name...");
        jPanel2.add(jTextField1, new org.netbeans.lib.awtextra.AbsoluteConstraints(90, 360, 300, -1));

        btnLogin1.setBackground(new java.awt.Color(255, 204, 51));
        btnLogin1.setText("Clear");
        btnLogin1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                btnLogin1ActionPerformed(evt);
            }
        });
        jPanel2.add(btnLogin1, new org.netbeans.lib.awtextra.AbsoluteConstraints(110, 670, 270, 50));

        jLabel9.setFont(new java.awt.Font("Verdana", 0, 15)); // NOI18N
        jLabel9.setForeground(new java.awt.Color(255, 255, 255));
        jLabel9.setText("Student ID:");
        jPanel2.add(jLabel9, new org.netbeans.lib.awtextra.AbsoluteConstraints(90, 430, -1, -1));

        jTextField2.setBackground(new java.awt.Color(153, 0, 0));
        jTextField2.setBorder(javax.swing.BorderFactory.createMatteBorder(0, 0, 2, 0, new java.awt.Color(255,
255, 255)));
        jTextField2.setFont(new java.awt.Font("Verdana", 0, 15)); // NOI18N
        jTextField2.setPlaceholder("Enter Student ID...");
        jPanel2.add(jTextField2, new org.netbeans.lib.awtextra.AbsoluteConstraints(90, 470, 300, -1));

        jLabel10.setFont(new java.awt.Font("Verdana", 0, 15)); // NOI18N
        jLabel10.setForeground(new java.awt.Color(255, 255, 255));
        jLabel10.setText("Student Name:");
        jPanel2.add(jLabel10, new org.netbeans.lib.awtextra.AbsoluteConstraints(90, 320, -1, -1));

        jLabel11.setFont(new java.awt.Font("Verdana", 0, 15)); // NOI18N
        jLabel11.setForeground(new java.awt.Color(255, 255, 255));
        jLabel11.setIcon(new
javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\icons8_Account_50px.png")); // NOI18N
        jPanel2.add(jLabel11, new org.netbeans.lib.awtextra.AbsoluteConstraints(30, 350, 50, 50));

        btnLogin2.setBackground(new java.awt.Color(255, 204, 51));
        btnLogin2.setText("Login");
        btnLogin2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                btnLogin2ActionPerformed(evt);
            }
        });
        jPanel2.add(btnLogin2, new org.netbeans.lib.awtextra.AbsoluteConstraints(110, 620, 270, 50));

        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE, 751,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
      layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
      .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE, 836, Short.MAX_VALUE)
      .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
    );

    setSize(new java.awt.Dimension(1220, 800));
    setLocationRelativeTo(null);
  }// </editor-fold>

  private void btnLogin1ActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    jTextField1.setText("");
    jTextField2.setText("");
  }

  private void btnLogin2ActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    String studentId = jTextField2.getText();
        String studentName = jTextField1.getText();

        String studentIdString = Integer.toString(Integer.parseInt(studentId));

        Library myLibrary = new Library();
        myLibrary.initializeStudents("C:\\Program Files\\NetBeansProjects\\students.txt");

        if(myLibrary.students.containsKey(studentIdString)){
          Student2 student = myLibrary.students.get(studentIdString);
          //Check if the studentName matches the name stored in the Student object
          if(student.getName().equals(studentName)) {
            JOptionPane.showMessageDialog(LoginAuthentication.this, "Welcome, Ga!\nYou have
successfully logged in.\n\n");

           BorrowReturn borrowReturn = new BorrowReturn(studentId,studentName);
            borrowReturn.setVisible(true);
              LoginAuthentication.this.dispose();
          }
        }else{
          JOptionPane.showMessageDialog(LoginAuthentication.this,"Login failed");
        }
  }

  private void jLabel5MouseClicked(java.awt.event.MouseEvent evt) {
    // TODO add your handling code here:
    System.exit(0);
  }

  /**
   * @param args the command line arguments
   */
  public static void main(String args[]) {
```

```
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look and feel.
         * For details see http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName());
                    break;
                }
            }
        } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(LoginAuthentication.class.getName()).log(java.util.logging.Level.SEVERE
, null, ex);
        } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(LoginAuthentication.class.getName()).log(java.util.logging.Level.SEVERE
, null, ex);
        } catch (IllegalAccessException ex) {

java.util.logging.Logger.getLogger(LoginAuthentication.class.getName()).log(java.util.logging.Level.SEVERE
, null, ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {

java.util.logging.Logger.getLogger(LoginAuthentication.class.getName()).log(java.util.logging.Level.SEVERE
, null, ex);
        }
        //</editor-fold>
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new LoginAuthentication().setVisible(true);
            }
        });
    }

    // Variables declaration - do not modify
    private rojerusan.RSMaterialButtonCircle btnLogin1;
    private rojerusan.RSMaterialButtonCircle btnLogin2;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel jLabel10;
    private javax.swing.JLabel jLabel11;
    private javax.swing.JLabel jLabel12;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel jLabel3;
    private javax.swing.JLabel jLabel4;
    private javax.swing.JLabel jLabel5;
    private javax.swing.JLabel jLabel6;
    private javax.swing.JLabel jLabel7;
    private javax.swing.JLabel jLabel8;
    private javax.swing.JLabel jLabel9;
```

```
    private javax.swing.JPanel jPanel1;
    private javax.swing.JPanel jPanel2;
    private app.bolivia.swing.JCTextField jTextField1;
    private app.bolivia.swing.JCTextField jTextField2;
    // End of variables declaration
}
```

Class for Borrow Return

```
/*
 * To change this license header, choose License Headers in Project Properties.
 * To change this template file, choose Tools | Templates
 * and open the template in the editor.
 */
package library.kiosk;

import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.ObjectInputStream;
import java.io.ObjectOutputStream;
import java.util.ArrayList;

/**
 *
 * @author Ninz
 */
public class BorrowReturn extends javax.swing.JFrame {

    /**
     * Creates new form BorrowReturn
     */
    public BorrowReturn(String studentId,String studentName) {
        this.studentId = studentId;
        this.studentName = studentName;
        initComponents();

        library2 = new Library();
        library2.initializeStudents("C:\\Program Files\\NetBeansProjects\\students.txt");

    }



    /**
     * This method is called from within the constructor to initialize the form.
     * WARNING: Do NOT modify this code. The content of this method is always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
```

```
        btnBorrow = new javax.swing.JButton();
        tfBookCode = new app.bolivia.swing.JCTextField();
        jLabel1 = new javax.swing.JLabel();
        jPanel3 = new javax.swing.JPanel();
        jScrollPane3 = new javax.swing.JScrollPane();
        bookTextArea = new javax.swing.JTextArea();
        btnReturn = new javax.swing.JButton();
        jLabel3 = new javax.swing.JLabel();
        jLabel4 = new javax.swing.JLabel();
        jLabel6 = new javax.swing.JLabel();
        jPanel2 = new javax.swing.JPanel();
        btnPrint = new javax.swing.JButton();
        jPanel4 = new javax.swing.JPanel();
        jScrollPane4 = new javax.swing.JScrollPane();
        jScrollPane1 = new javax.swing.JScrollPane();
        receipt = new javax.swing.JTextArea();
        jLabel2 = new javax.swing.JLabel();
        btnClear = new rojerusan.RSMaterialButtonCircle();
        btnClosed = new rojerusan.RSMaterialButtonCircle();
        jLabel5 = new javax.swing.JLabel();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setBackground(new java.awt.Color(255, 255, 255));
        setUndecorated(true);
        setType(java.awt.Window.Type.POPUP);
        addWindowListener(new java.awt.event.WindowAdapter() {
            public void windowClosing(java.awt.event.WindowEvent evt) {
                formWindowClosing(evt);
            }
            public void windowOpened(java.awt.event.WindowEvent evt) {
                formWindowOpened(evt);
            }
        });

        jPanel1.setBackground(new java.awt.Color(153, 0, 0));
        jPanel1.setCursor(new java.awt.Cursor(java.awt.Cursor.DEFAULT_CURSOR));
        jPanel1.setLayout(new org.netbeans.lib.awtextra.AbsoluteLayout());

        btnBorrow.setBackground(new java.awt.Color(255, 255, 255));
        btnBorrow.setFont(new java.awt.Font("Verdana", 1, 15)); // NOI18N
        btnBorrow.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\book.png")); // NOI18N
        btnBorrow.setText("BORROW");
        btnBorrow.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                btnBorrowActionPerformed(evt);
            }
        });
        jPanel1.add(btnBorrow, new org.netbeans.lib.awtextra.AbsoluteConstraints(120, 840, 220, 80));

        tfBookCode.setBackground(new java.awt.Color(153, 0, 0));
        tfBookCode.setBorder(javax.swing.BorderFactory.createMatteBorder(0, 0, 2, 0, new
java.awt.Color(255, 255, 255)));
        tfBookCode.setHorizontalAlignment(javax.swing.JTextField.CENTER);
        tfBookCode.setFont(new java.awt.Font("Verdana", 1, 25)); // NOI18N
        tfBookCode.setPlaceholder("Enter book code...");
        jPanel1.add(tfBookCode, new org.netbeans.lib.awtextra.AbsoluteConstraints(410, 750, 310, 50));
```

```
jLabel1.setFont(new java.awt.Font("Stencil", 1, 40)); // NOI18N
jLabel1.setForeground(new java.awt.Color(255, 204, 51));
jLabel1.setText("Book Code");
jPanel1.add(jLabel1, new org.netbeans.lib.awtextra.AbsoluteConstraints(450, 710, -1, -1));

bookTextArea.setColumns(20);
bookTextArea.setFont(new java.awt.Font("Verdana", 0, 18)); // NOI18N
bookTextArea.setRows(5);
jScrollPane3.setViewportView(bookTextArea);

javax.swing.GroupLayout jPanel3Layout = new javax.swing.GroupLayout(jPanel3);
jPanel3.setLayout(jPanel3Layout);
jPanel3Layout.setHorizontalGroup(
   jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
   .addComponent(jScrollPane3, javax.swing.GroupLayout.DEFAULT_SIZE, 1130, Short.MAX_VALUE)
);
jPanel3Layout.setVerticalGroup(
   jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
   .addComponent(jScrollPane3, javax.swing.GroupLayout.DEFAULT_SIZE, 490, Short.MAX_VALUE)
);

jPanel1.add(jPanel3, new org.netbeans.lib.awtextra.AbsoluteConstraints(20, 90, 1130, 490));

btnReturn.setBackground(new java.awt.Color(255, 255, 255));
btnReturn.setFont(new java.awt.Font("Verdana", 1, 15)); // NOI18N
btnReturn.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\book (1).png")); // NOI18N
btnReturn.setText(" RETURN");
btnReturn.addActionListener(new java.awt.event.ActionListener() {
   public void actionPerformed(java.awt.event.ActionEvent evt) {
      btnReturnActionPerformed(evt);
   }
});
jPanel1.add(btnReturn, new org.netbeans.lib.awtextra.AbsoluteConstraints(800, 840, 220, 80));

jLabel3.setFont(new java.awt.Font("SimSun-ExtB", 1, 30)); // NOI18N
jLabel3.setForeground(new java.awt.Color(255, 255, 255));
jLabel3.setText("AVAILABLE BOOKS");
jPanel1.add(jLabel3, new org.netbeans.lib.awtextra.AbsoluteConstraints(440, 590, -1, -1));

jLabel4.setFont(new java.awt.Font("Verdana", 1, 20)); // NOI18N
jLabel4.setForeground(new java.awt.Color(255, 204, 51));
jLabel4.setIcon(new javax.swing.ImageIcon("C:\\Program Files\\NetBeansProjects\\Mini Library
Kiosk\\src\\Icon\\imresizer-1709272030013.png")); // NOI18N
jLabel4.setText(" WIBRARY");
jPanel1.add(jLabel4, new org.netbeans.lib.awtextra.AbsoluteConstraints(0, 0, 250, 80));

jLabel6.setFont(new java.awt.Font("Tahoma", 0, 13)); // NOI18N
jLabel6.setForeground(new java.awt.Color(255, 255, 255));
jLabel6.setText("Knowledge at Your Fingertips");
jPanel1.add(jLabel6, new org.netbeans.lib.awtextra.AbsoluteConstraints(130, 50, 170, 20));

jPanel2.setBackground(new java.awt.Color(255, 204, 51));

btnPrint.setBackground(new java.awt.Color(255, 255, 255));
btnPrint.setFont(new java.awt.Font("Verdana", 1, 15)); // NOI18N
```

```
        btnPrint.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\print.png")); // NOI18N
        btnPrint.setText(" PRINT");
        btnPrint.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                btnPrintActionPerformed(evt);
            }
        });

        receipt.setColumns(20);
        receipt.setFont(new java.awt.Font("Monospaced", 1, 15)); // NOI18N
        receipt.setRows(5);
        jScrollPane1.setViewportView(receipt);

        jScrollPane4.setViewportView(jScrollPane1);

        javax.swing.GroupLayout jPanel4Layout = new javax.swing.GroupLayout(jPanel4);
        jPanel4.setLayout(jPanel4Layout);
        jPanel4Layout.setHorizontalGroup(
            jPanel4Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jScrollPane4)
        );
        jPanel4Layout.setVerticalGroup(
            jPanel4Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel4Layout.createSequentialGroup()
                .addComponent(jScrollPane4, javax.swing.GroupLayout.PREFERRED_SIZE, 494,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(0, 0, Short.MAX_VALUE))
        );

        jLabel2.setFont(new java.awt.Font("Tahoma", 1, 13)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(255, 255, 255));
        jLabel2.setIcon(new javax.swing.ImageIcon("C:\\Program Files\\NetBeansProjects\\Mini Library
Kiosk\\src\\Icon\\imresizer-1709050662631.png")); // NOI18N
        jLabel2.setText("UNIVERSITY OF MINDANAO");

        btnClear.setBackground(new java.awt.Color(153, 0, 0));
        btnClear.setText("CLEAR");
        btnClear.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                btnClearActionPerformed(evt);
            }
        });

        btnClosed.setBackground(new java.awt.Color(153, 0, 0));
        btnClosed.setText("CLOSED");
        btnClosed.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                btnClosedActionPerformed(evt);
            }
        });

        jLabel5.setFont(new java.awt.Font("Tw Cen MT Condensed Extra Bold", 1, 18)); // NOI18N
        jLabel5.setIcon(new javax.swing.ImageIcon("C:\\Users\\Ninz\\Downloads\\close-window.png")); //
NOI18N
        jLabel5.addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseClicked(java.awt.event.MouseEvent evt) {
```

```
            jLabel5MouseClicked(evt);
         }
      });

      javax.swing.GroupLayout jPanel2Layout = new javax.swing.GroupLayout(jPanel2);
      jPanel2.setLayout(jPanel2Layout);
      jPanel2Layout.setHorizontalGroup(
         jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
         .addGroup(jPanel2Layout.createSequentialGroup()
            .addContainerGap()
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
               .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanel2Layout.createSequentialGroup()
                  .addComponent(jPanel4, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                  .addContainerGap())
               .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanel2Layout.createSequentialGroup()
                  .addGap(0, 0, Short.MAX_VALUE)
                  .addComponent(btnPrint, javax.swing.GroupLayout.PREFERRED_SIZE, 220,
javax.swing.GroupLayout.PREFERRED_SIZE)
                  .addGap(95, 95, 95))))
         .addGroup(jPanel2Layout.createSequentialGroup()
            .addComponent(jLabel2)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addComponent(jLabel5))
         .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, jPanel2Layout.createSequentialGroup()
            .addContainerGap(44, Short.MAX_VALUE)
            .addComponent(btnClear, javax.swing.GroupLayout.PREFERRED_SIZE, 156,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(38, 38, 38)
            .addComponent(btnClosed, javax.swing.GroupLayout.PREFERRED_SIZE, 156,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(39, 39, 39))
      );
      jPanel2Layout.setVerticalGroup(
         jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
         .addGroup(jPanel2Layout.createSequentialGroup()
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
               .addComponent(jLabel2)
               .addComponent(jLabel5, javax.swing.GroupLayout.PREFERRED_SIZE, 32,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(1, 1, 1)
            .addComponent(jPanel4, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(18, 18, 18)
            .addComponent(btnPrint, javax.swing.GroupLayout.PREFERRED_SIZE, 80,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
               .addComponent(btnClosed, javax.swing.GroupLayout.PREFERRED_SIZE, 73,
javax.swing.GroupLayout.PREFERRED_SIZE)
               .addComponent(btnClear, javax.swing.GroupLayout.PREFERRED_SIZE, 73,
javax.swing.GroupLayout.PREFERRED_SIZE))
```

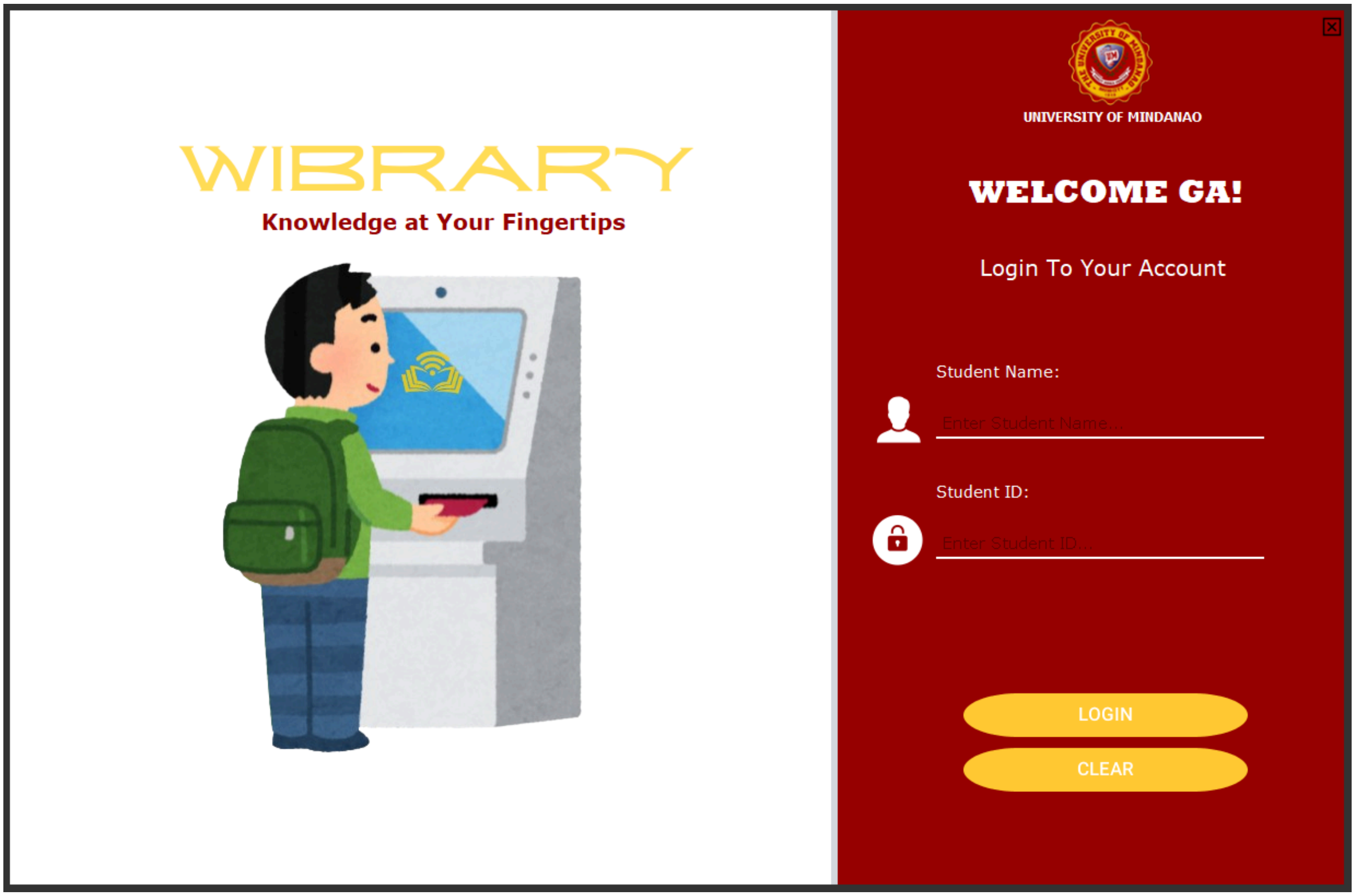
```
            .addGap(93, 93, 93))
        );

        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE, 1167, Short.MAX_VALUE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE, 1010, Short.MAX_VALUE)
            .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
        );

        pack();
        setLocationRelativeTo(null);
    }// </editor-fold>

    private void btnBorrowActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        library2.borrowBook(tfBookCode.getText(),studentId,studentName, bookTextArea);

        receipt.setText("");
        tfBookCode.setText("");
    }

    private void btnReturnActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        library2.returnBook(tfBookCode.getText(),studentId,studentName,bookTextArea);

        receipt.setText("");
        tfBookCode.setText("");
    }

    private void btnPrintActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        library2.displayReceipt(receipt);
    }

    private void formWindowOpened(java.awt.event.WindowEvent evt) {
        // TODO add your handling code here:
         try {
        FileInputStream file = new FileInputStream("file.bin");
        ObjectInputStream input = new ObjectInputStream(file);


        // Read the serialized ArrayList<Book2> from the file
        ArrayList<Book2> booksFromFile = (ArrayList<Book2>) input.readObject();
```

```
        input.close();
        file.close();

        // Set the retrieved ArrayList<Book2> back to the books field in your Library class
        library2.books = booksFromFile;

        library2.displayBooks2(bookTextArea);
        library2.loadBorrowersFromFile();


    } catch (Exception ex) {
        ex.printStackTrace();


    }
    }

    private void formWindowClosing(java.awt.event.WindowEvent evt) {
        // TODO add your handling code here:
        library2.loadBooksFromFile("C:\\Program Files\\NetBeansProjects\\books.txt");


        try {
        FileOutputStream file = new FileOutputStream("file.bin");
        ObjectOutputStream output = new ObjectOutputStream(file);


        // Use the getBooks() method to retrieve the books ArrayList
        output.writeObject(library2.getBooks()); // Writing the books ArrayList to file

        output.close();
        file.close();


    } catch (Exception ex) {
        ex.printStackTrace();
    }
    }

    private void jLabel5MouseClicked(java.awt.event.MouseEvent evt) {
        // TODO add your handling code here:
        System.exit(0);
    }

    private void btnClearActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        tfBookCode.setText("");
        receipt.setText("");
    }

    private void btnClosedActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
          library2.loadBooksFromFile("book2.txt");

        try {
        FileOutputStream file = new FileOutputStream("file.bin");
```

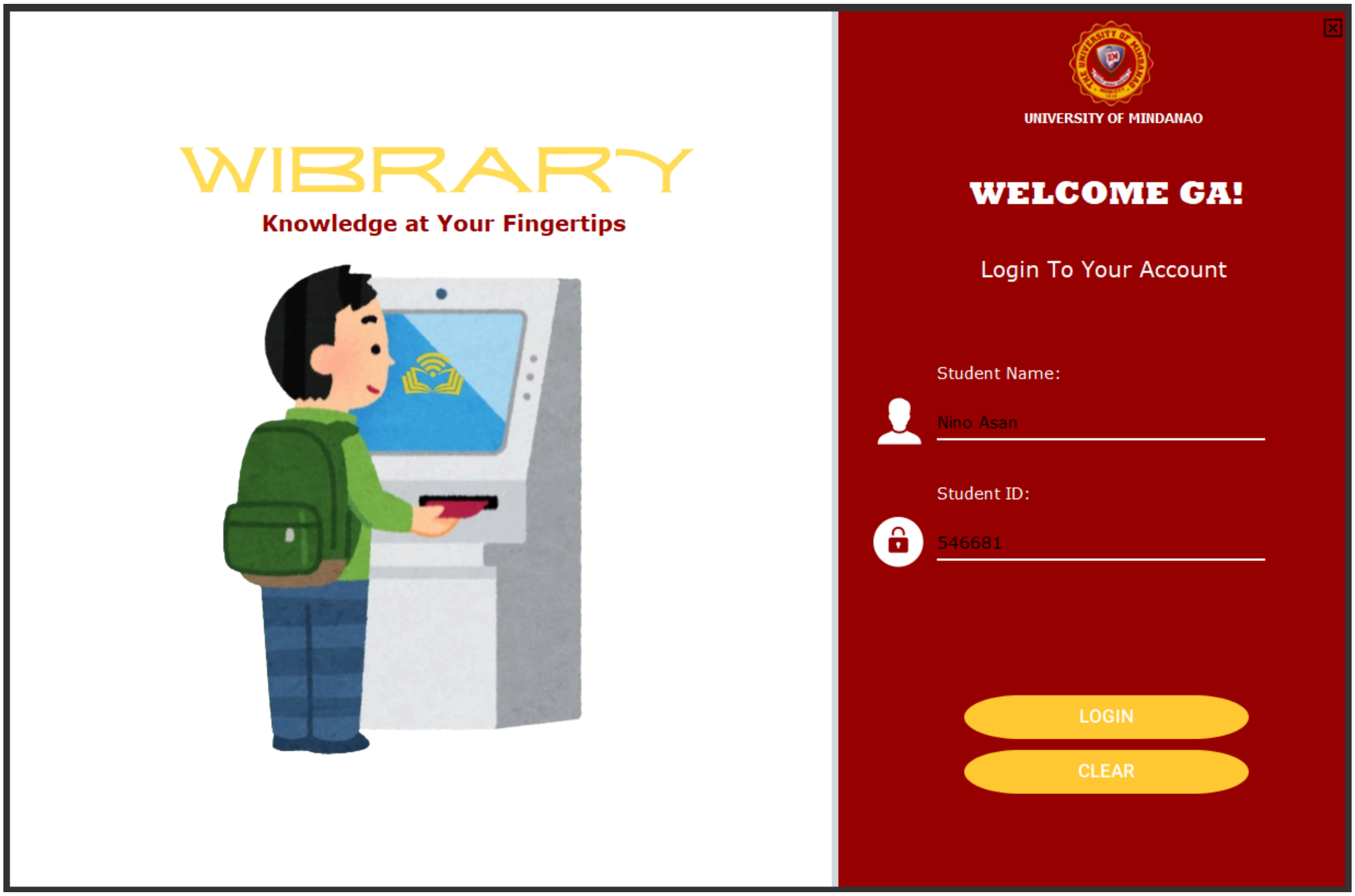
```
        ObjectOutputStream output = new ObjectOutputStream(file);

        output.writeObject(library2.getBooks());

        output.close();
        file.close();


    } catch (Exception ex) {
        ex.printStackTrace();
    }
        this.dispose();
    }

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look and feel.
         * For details see http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName());
                    break;
                }
            }
        } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(BorrowReturn.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
        } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(BorrowReturn.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
        } catch (IllegalAccessException ex) {

java.util.logging.Logger.getLogger(BorrowReturn.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {

java.util.logging.Logger.getLogger(BorrowReturn.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
        }
        //</editor-fold>

        /* Create and display the form */
         java.awt.EventQueue.invokeLater(new Runnable() {
            String studentId = ""; // Initialize with an empty string
            String studentName = ""; // Initialize with an empty string

            public void run() {
```

```
            new BorrowReturn(studentId,studentName);//.setVisible(true);
        }
    });
}
private Library library2;
private Library library;
private String studentId;
private String studentName;
// Variables declaration - do not modify
private javax.swing.JTextArea bookTextArea;
private javax.swing.JButton btnBorrow;
private rojerusan.RSMaterialButtonCircle btnClear;
private rojerusan.RSMaterialButtonCircle btnClosed;
private javax.swing.JButton btnPrint;
private javax.swing.JButton btnReturn;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JLabel jLabel6;
private javax.swing.JPanel jPanel1;
private javax.swing.JPanel jPanel2;
private javax.swing.JPanel jPanel3;
private javax.swing.JPanel jPanel4;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JScrollPane jScrollPane3;
private javax.swing.JScrollPane jScrollPane4;
private javax.swing.JTextArea receipt;
private app.bolivia.swing.JCTextField tfBookCode;
// End of variables declaration
}
```

VII. System Output



This is our display for login authentication. It features two (2) text fields for the student ID and name, as well as two (2) buttons that let you clear the text fields (Clear) and log in to the library (Login).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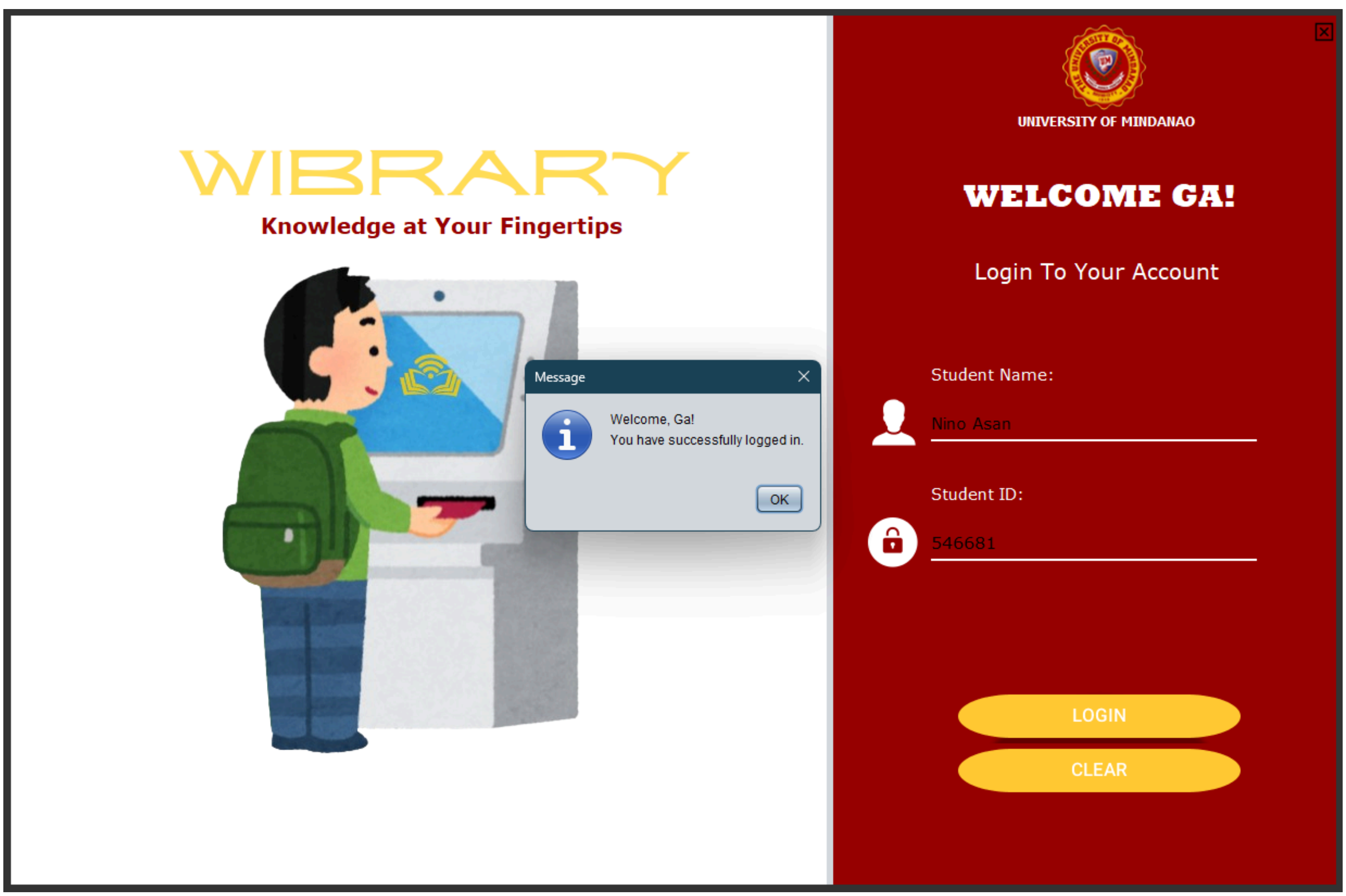
WELCOME GA!
Login To Your Account
Student Name:
Nino Asan
Student ID:
546681
LOGIN
CLEAR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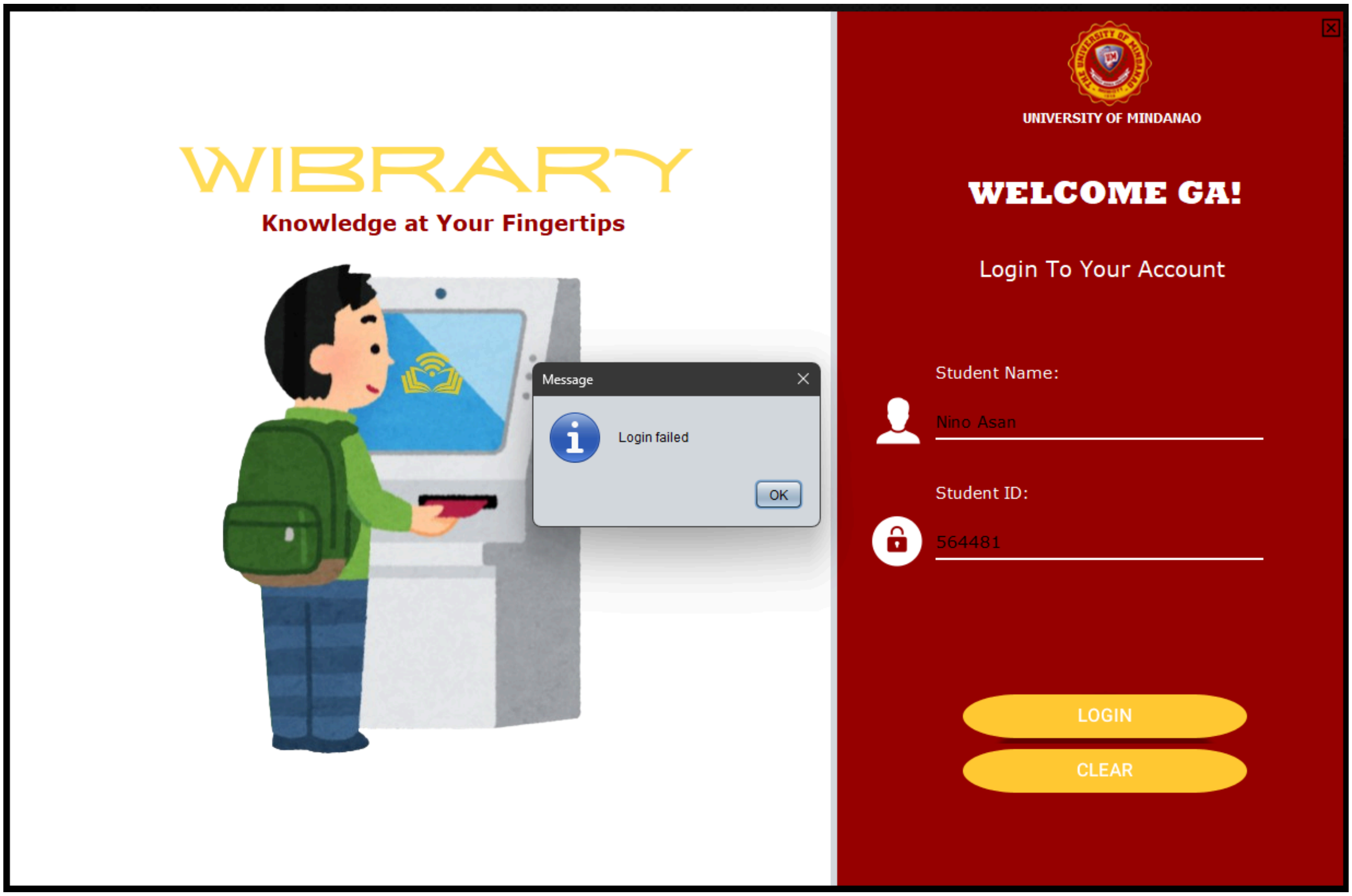
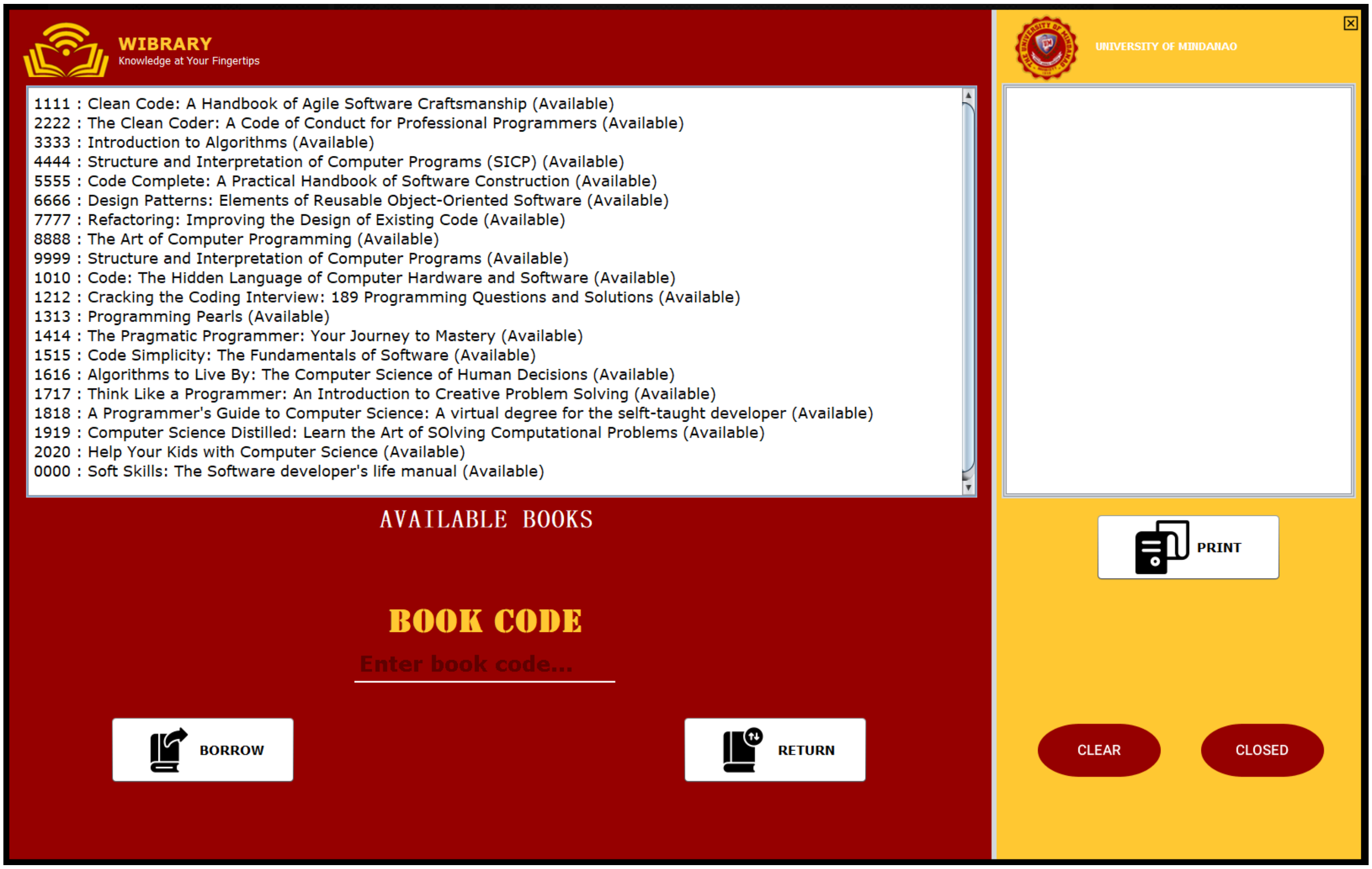
Message
Welcome, Ga!
You have successfully logged in.
OK
UNIVERSITY OF MINDANAO
WELCOME GA!
Login To Your Account
Student Name:
Nino Asan
Student ID:
546681
LOGIN
CLEAR



To access the library, you must first enter your student ID and name in the appropriate text areas before clicking the "Login" button. There's a "Clear" button to clear the text fields and enter the details again if something goes wrong. Then a notification pops out to inform you that your login attempt was successful.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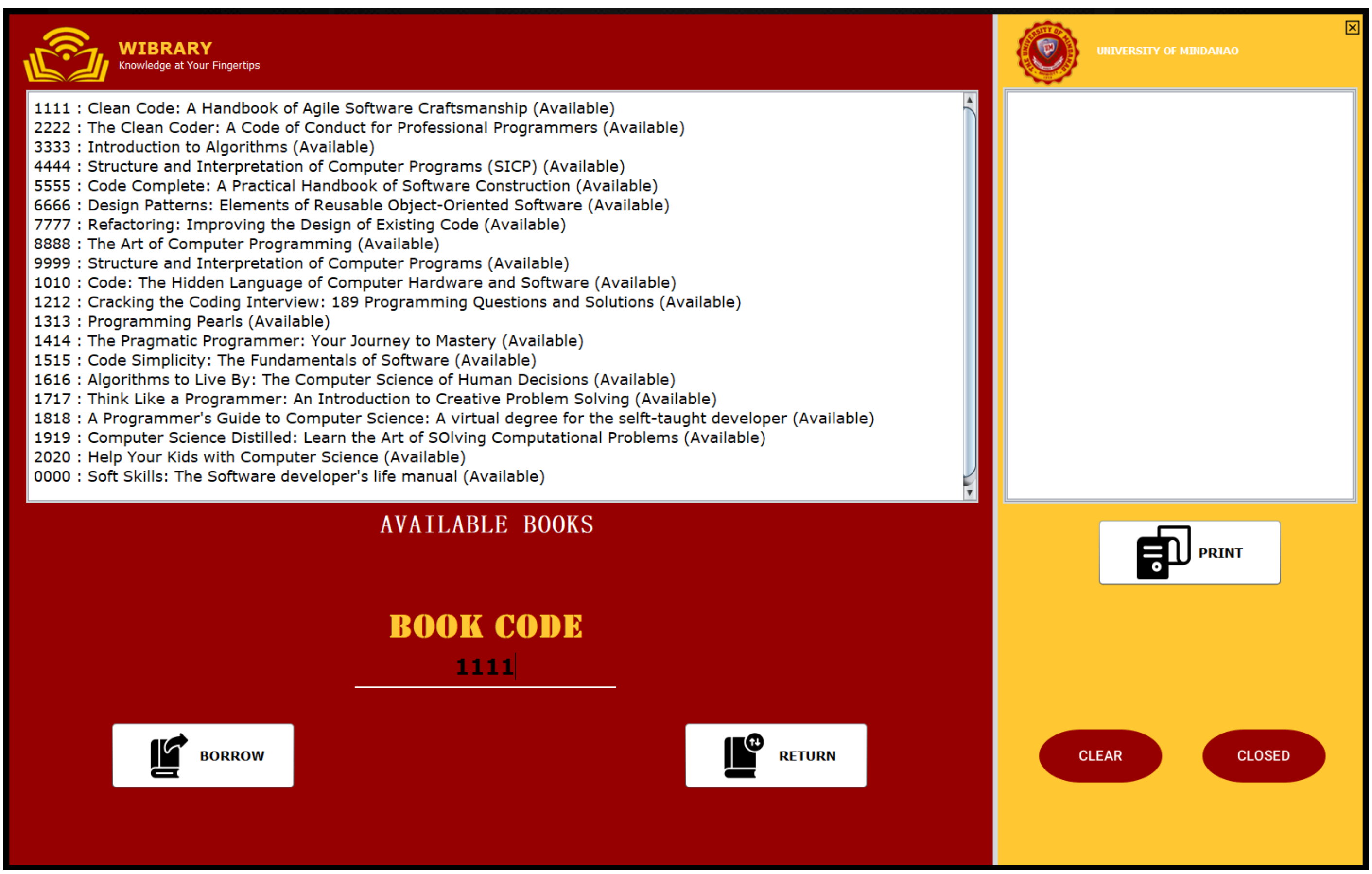
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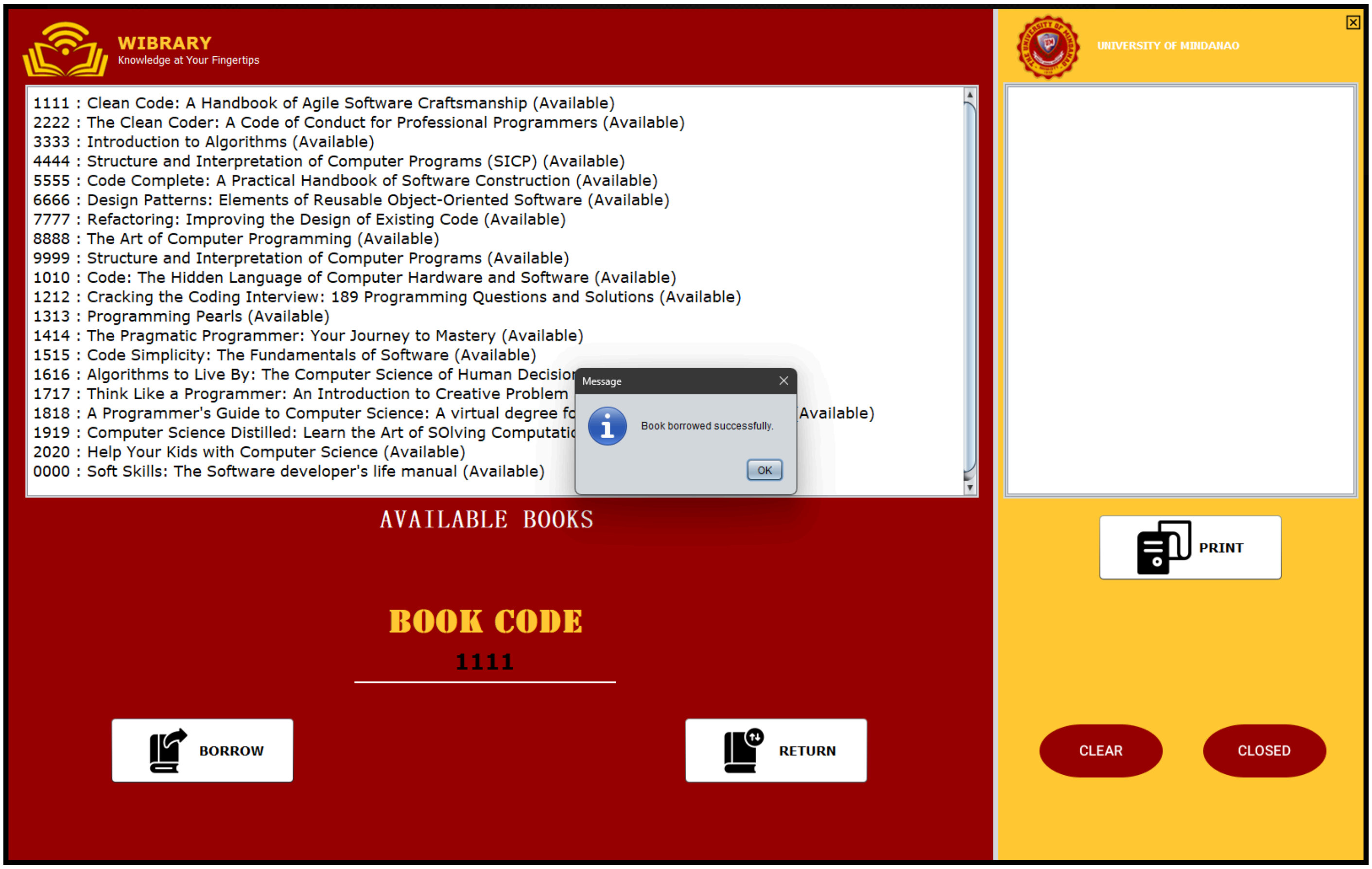
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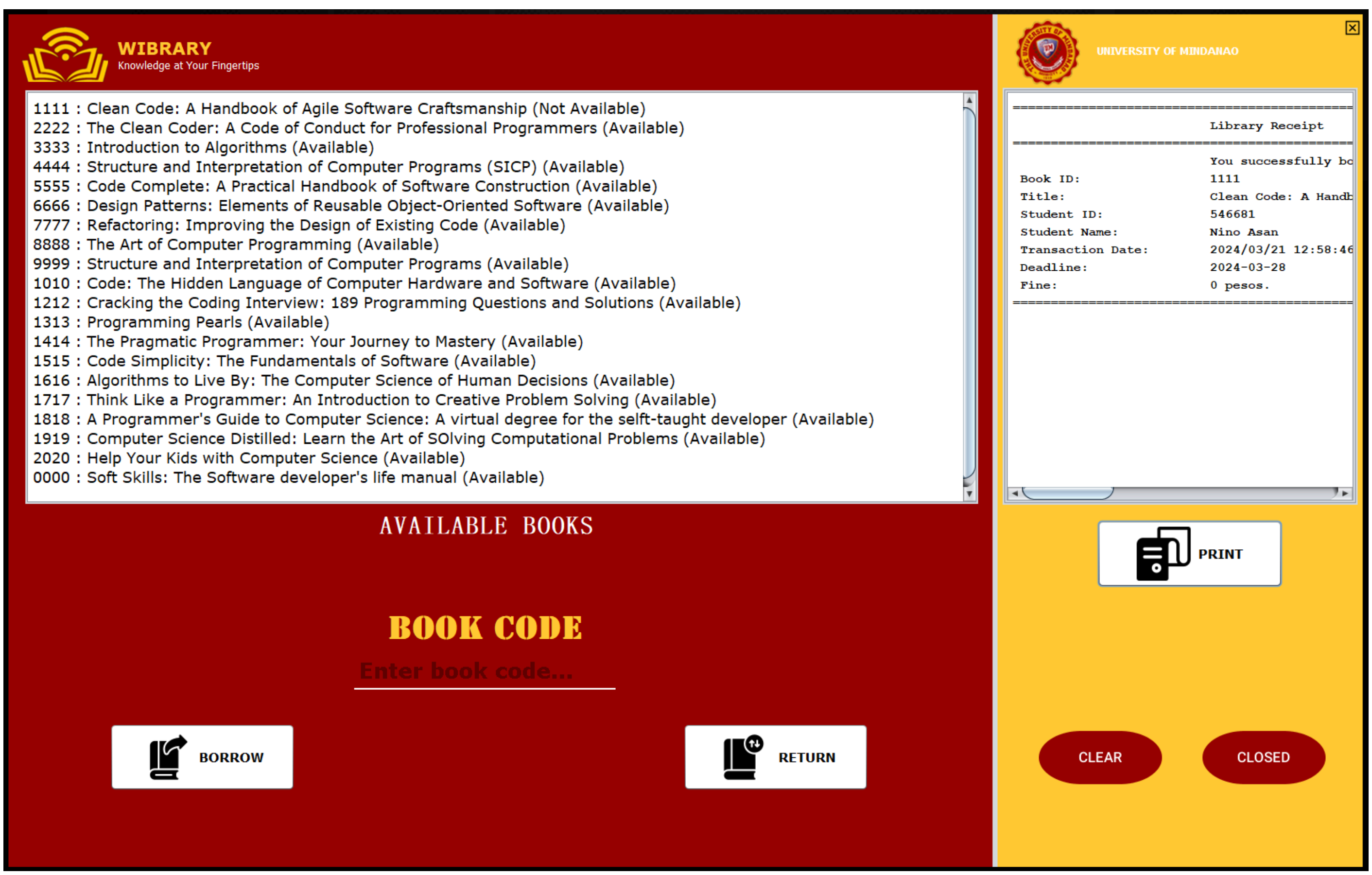
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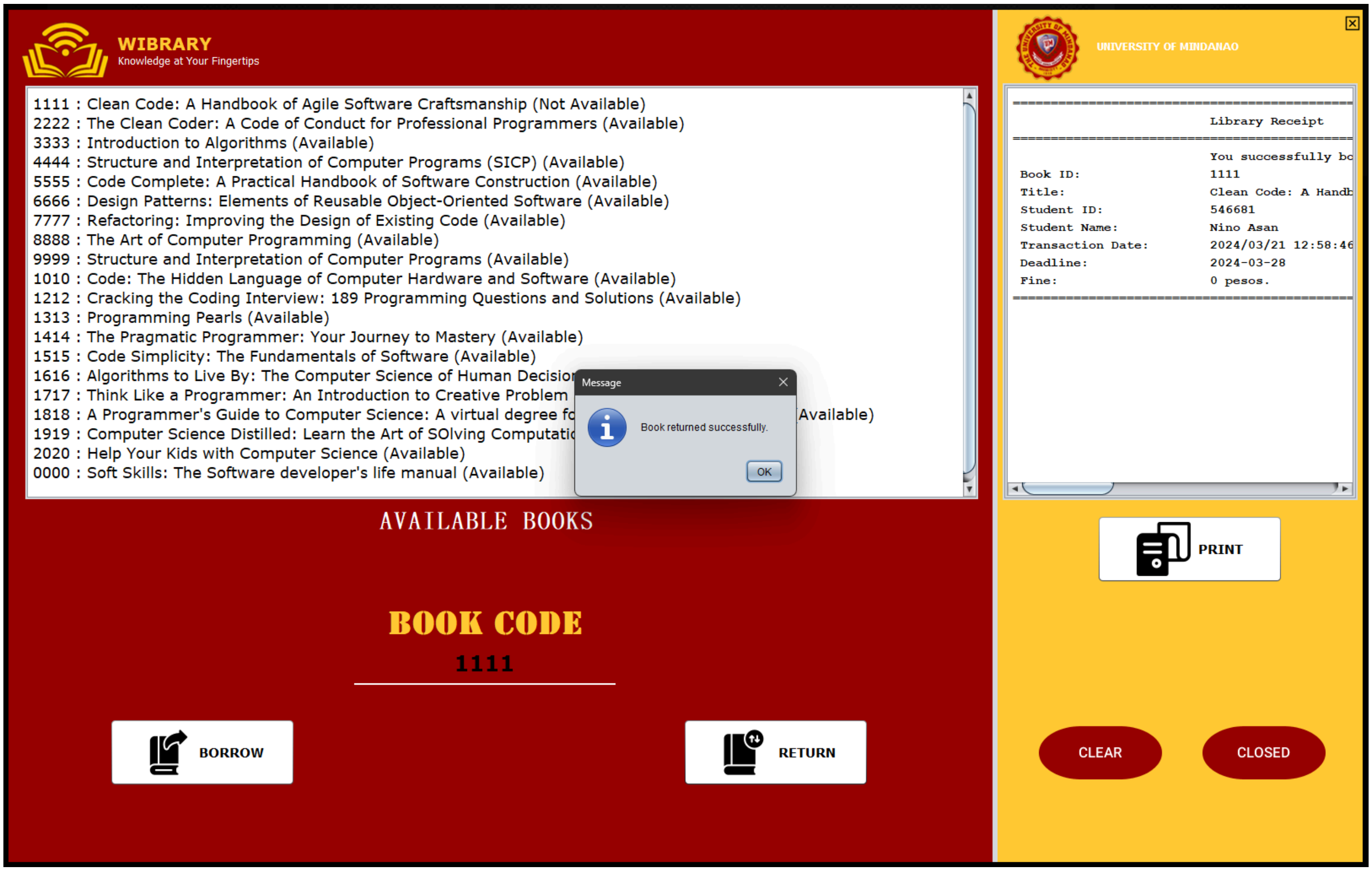
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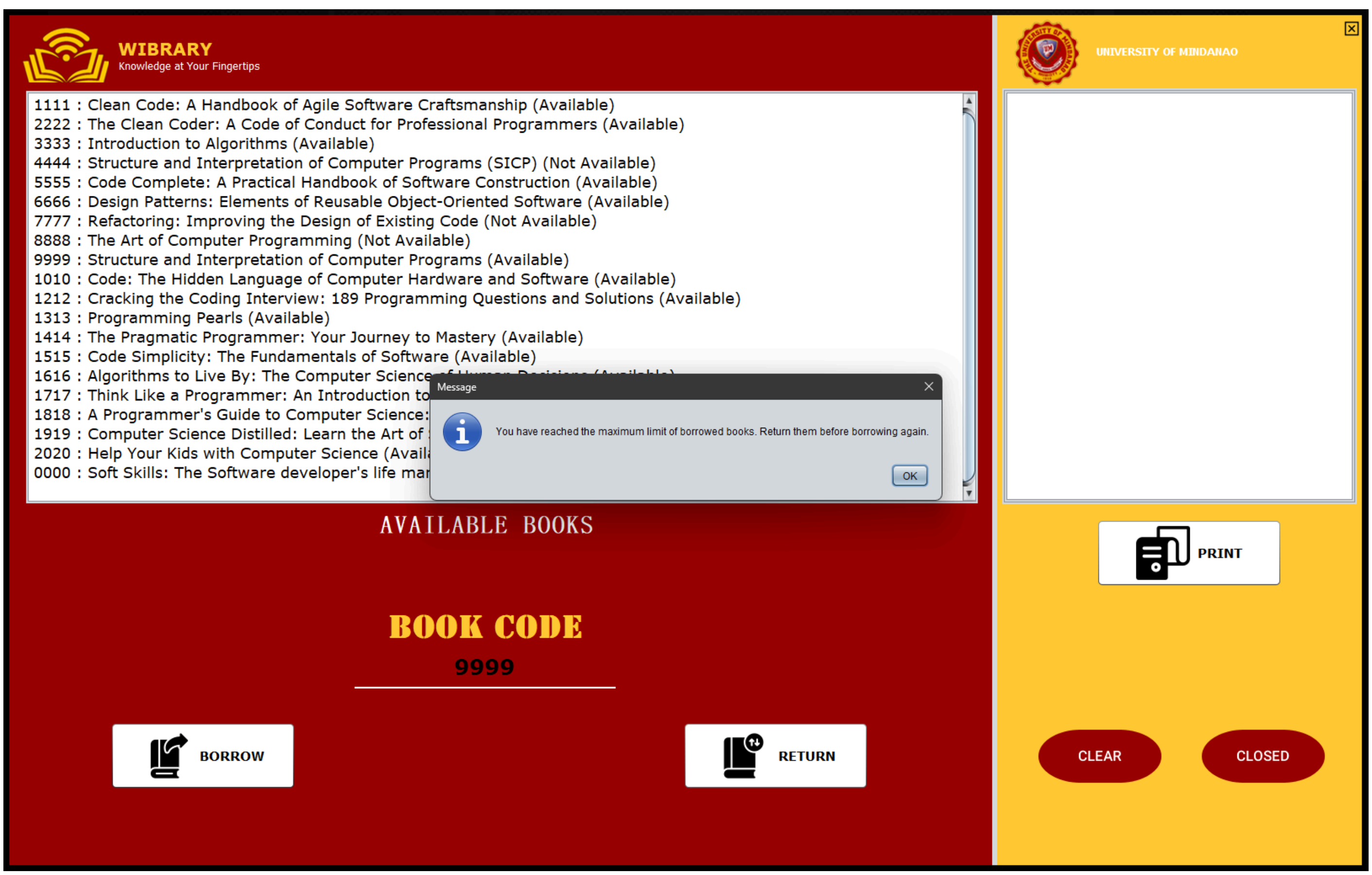
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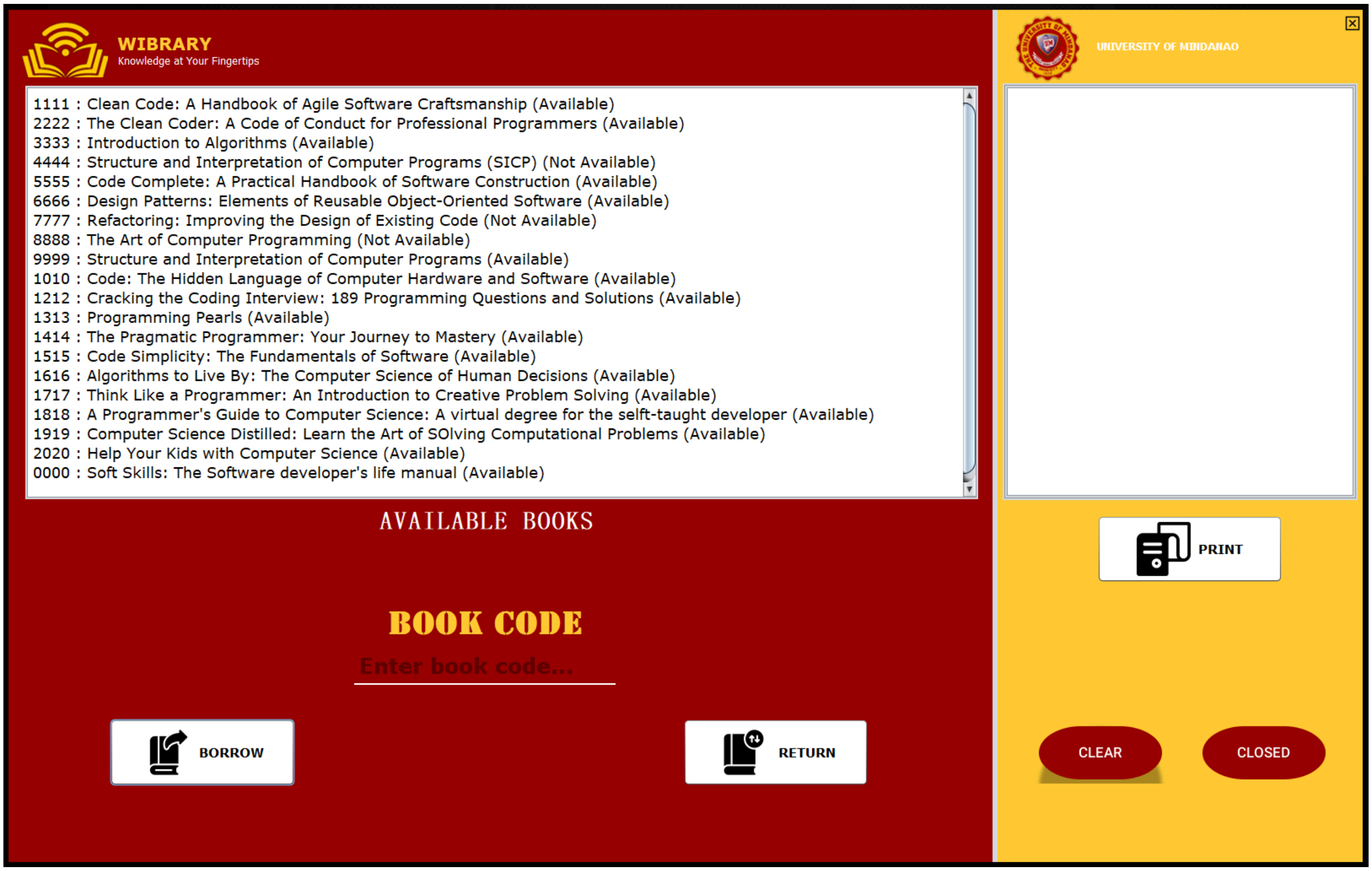
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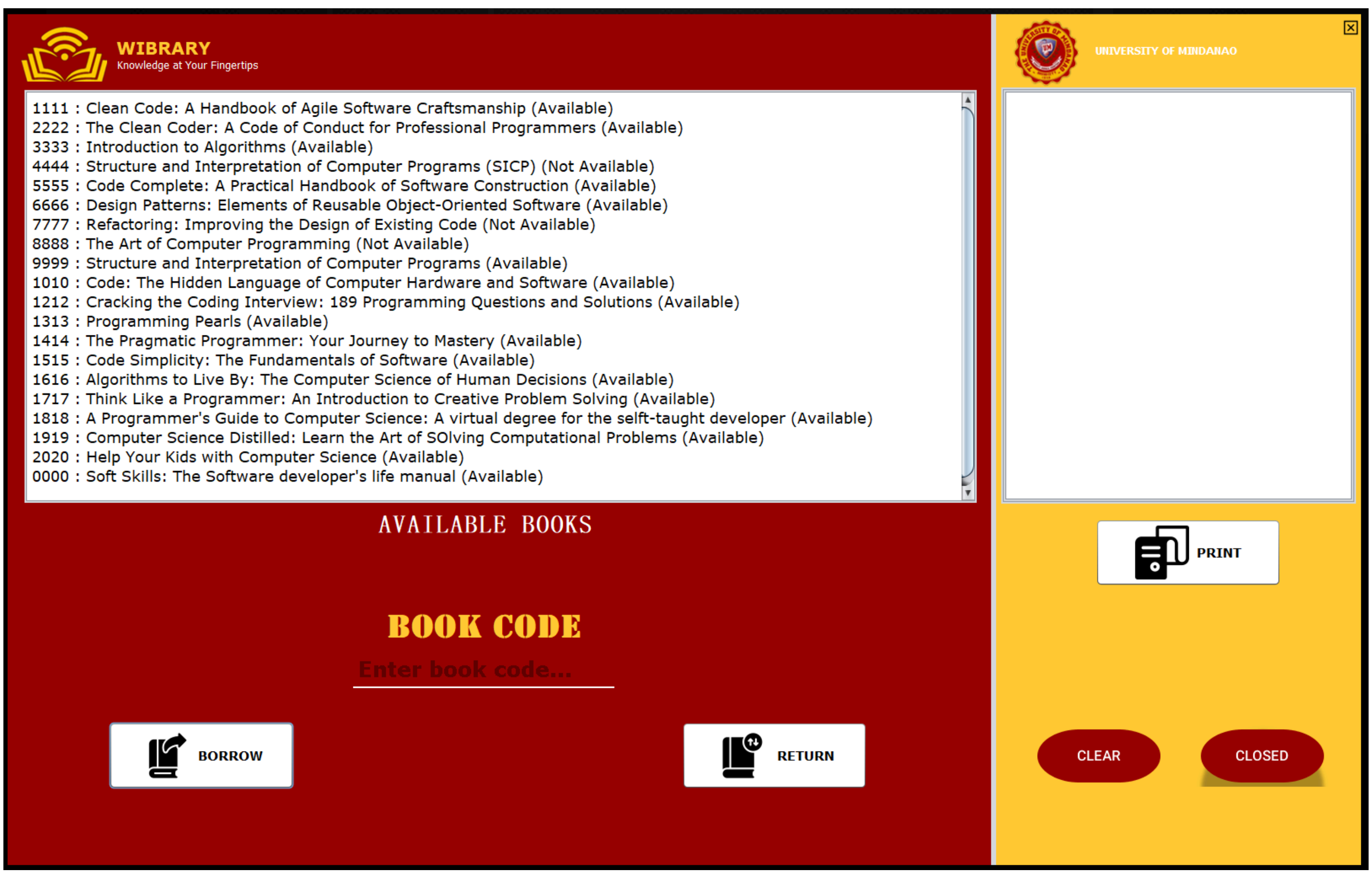
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We have the library as our second interface. It features a book list section, a receipt area, one (1) text field for the book code needed to borrow and return books, as well as four (4) buttons. Click the “Borrow” button to borrow books from the list, then click the “Return” button to return the books that were borrowed, Click the “Print” button to print the receipt, where the important details are listed, including the date you borrowed the books and the date you had to return them to avoid penalties, as well as the penalty fee, which is also stated. To remove all of the text fields, click the "Clear" button. Finally, a notification pops out to inform you that you borrowed and returned the books successfully, as well as to inform you that you reached the limit on borrowing books.